When Does Temperature Scaling Pay Off in Cross-Corpus ECG Transfer? A Multi-Seed Calibration Boundary Study

Toni Guan[a,*]

[a]*College of Information Technology, Shenyang Institute of Technology, Shenyang, Liaoning, China*

Abstract

Temperature scaling (TS) is the default post-hoc recalibration for deep classifiers, yet whether its in-distribution (ID) benefit transfers to out-of-distribution (OOD) target corpora remains the central open question for cross-institution ECG deployment. We report a multi-seed calibration boundary study on six directed transfer pairs among Chapman–Shaoxing, CPSC2018, and PTB-XL, with two architectures (InceptionTime and ResNet1D), five seeds (42–46), eight recalibration methods, and a patient-level cluster paired bootstrap (BCa, $B = 10{,}000$) on the full 60-experiment grid. The study followed a pre-registered protocol with dated, publicly archived amendments. The primary endpoint is the OOD calibration benefit $\Delta\mathrm{ECE}_{\mathrm{OOD}} = \mathrm{ECE}^{\mathrm{OOD}}_{\mathrm{raw}} - \mathrm{ECE}^{\mathrm{OOD}}_{\mathrm{TS}}$, tested one-sided against a pre-specified direction prior. TS yields a positive OOD benefit in 51/60 seed experiments (85.0%; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples disclosed and mechanistically interpreted; excluding six degenerate-CI cells gives 45/60 (75.0%). The pre-specified ID boundary criterion ($\geq$80% of cells with 95% CIs containing zero) is *not* met (23/60 = 38.3%), triggering the registered "ID nonzero boundary" branch: the OOD benefit is about 1.9$\times$ the ID benefit ($+0.0159$ vs.

*Corresponding author.

+0.0083). A validated 27-cell Shapley decomposition separates slope, intercept, and prevalence effects; a predictability experiment triggers the registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, although a source-fitted, label-free temperature predicts the *sign* of the benefit in 60/60 cells); and a discrimination-aware gate (C5) shows 7/12 pair$\times$architecture strata passing both calibration and discrimination layers. Together these results bound when post-hoc recalibration pays off across institutions, and when it does not.

Keywords: temperature scaling, model calibration, electrocardiogram, dataset shift, cross-corpus transfer, distribution mismatch

---

1. Introduction

Modern ECG deep models achieve strong in-distribution (ID) accuracy and calibration, but their reliability under distribution shift remains open [1, 2, 3]. Recalibration fits a small parameter set on a labeled calibration split; its benefit, a reduction in expected calibration error (ECE), is measured on that same distribution and reverses on a shifted target.

For ECG this question is not academic. PTB-XL [3], Chapman-Shaoxing [4], and CPSC use different acquisition protocols, lead sets, and label vocabularies, so a model calibrated at one institution meets covariate, prior-probability, and label-concept shift at the next [25]. Clinicians and regulators then face a concrete decision: *given my local data budget and the measurable shift, should I deploy the recalibrated model, buy labels, or refuse?*

Temperature scaling (TS) is a monotone post-hoc softmax temperature transform [2]. Fitted under a convex loss, it minimizes NLL on the calibration split, so its calibration benefit there is empirically small; on a shifted target

the source-fit temperature is mismatched to the target logits and a non-trivial benefit can remain. This asymmetry is *empirical*, not a theorem: ECE is not invariant under confidence-shifting monotone maps, and TS can *increase* ECE where the model is under-confident (nine negative cells in our data). It motivates the pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ over the originally planned decay $\Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (revision A1, Sec. 4.4). To avoid HARKing, protocol v2.1-A1 [26] fixes the primary endpoint, test, family structure, and failure branches before unblinding; revision A1 re-locates the endpoint from `decay` to $\Delta\text{ECE}_{\text{OOD}}$, triggered by an independent pilot, and is archived with before/after protocol hashes [27, 28]. Multi-seed validation (5 seeds $\times$ 6 transfer pairs $\times$ 2 architectures) supplies the robustness evidence; all 60 seed results, including nine counter-examples, are reported.

Five independent contributions:

C1 (Primary OOD benefit): On the pre-registered endpoint $\Delta\text{ECE}_{\text{OOD}}$, TS yields a statistically significant positive OOD benefit in 51/60 seed experiments (6 transfer pairs $\times$ 2 architectures; InceptionTime 27/30, ResNet1D 24/30), with nine counter-examples transparently disclosed and mechanistically explained. InceptionTime: three of six directions achieve 5/5, three achieve 4/5. ResNet1D: two of six achieve 5/5, two 4/5, two 3/5.

C2 (Boundary condition, pre-registered branch triggered): Across 12 pair$\times$architecture strata, 23/60 cells (38.3%; BCa operating CI) have 95% CIs containing zero and 34/60 have CI lower bound > 0 (stratum means -0.0008 to $+0.0232$; ID mean $+0.0083$ vs. OOD mean $+0.0159$). The pre-registered criterion ($\geq$80% of cells with CI containing zero) is *not* met, triggering the "ID nonzero boundary" branch (protocol A1.4.2): the

premise that TS has no ID recalibratable miscalibration is re-examined, the decay contrast restored as exploratory, and C1 qualified, since the OOD benefit is about $1.9\times$ the ID benefit but is *not* exclusively OOD-specific. $47/60 = 78.3\%$ of ID point estimates are positive and $2/60 = 3.3\%$ have $\Delta\text{ECE}_{\text{ID}} < -0.01$ (counter-example 6, -0.0152; one marginal case, -0.0128).

C3 (Methodological, Shapley decomposition validation): A 27-cell full-factorial synthetic validation (slope$\times$intercept$\times$prevalence, $n = 20{,}000$) recovers ground truth in the identifiable regime (27/27 at $n = 20{,}000$; 21/27 at $n = 500$), flags entangled regions without a threshold, and reports bootstrap CIs on absolute Shapley values.

C4 (Practical, method-selection boundary): Across 8 methods $\times$ 6 pairs $\times$ 5 seeds $\times$ 2 architectures, TS is robust (51/60) while EM prior adaptation is fragile when ill-conditioned cells ($w_{\text{ID}} > 10$) are included (InceptionTime 15/30); after exclusion the EM OOD finding attenuates. The mechanism is prior-likelihood mismatch: EM fits a target prior that amplifies shift-induced miscalibration when the source confusion matrix is ill-conditioned.

C5 (Clinical, discrimination-aware calibration gate): A two-layer gate decouples calibration effectiveness (C1: $\Delta\text{ECE}_{\text{OOD}} > 0$) from clinical viability (accuracy $\geq$ source-domain baseline). Across 12 pair$\times$architecture strata, 7/12 pass both layers; 5/12 fail the discrimination layer, including directions with positive $\Delta\text{ECE}_{\text{OOD}}$ but accuracy below baseline (e.g., PTB-XL$\rightarrow$Chapman ResNet1D: $\Delta\text{ECE} = +0.029$, acc $= 0.657 < 0.713$). The

gate reframes the 9/60 counter-examples: 4/9 calibration failures (C1 rejects) and 5/9 discrimination failures (C5 rejects despite C1 passing).

Prior ECG calibration studies report the phenomenon [12, 13, 20, 21] but not an actionable *when does TS pay off* answer with multi-seed robustness evidence. The novelty is *not* multi-seed validation on ECG (a methodological choice) but the **first systematic characterization of the boundary conditions under which TS yields a positive OOD calibration benefit in cross-corpus ECG transfer**: the pre-registered OOD quantification, the ID boundary condition, the method-selection boundary (TS robust; EM prior and Matrix scaling weakest), the predictability failure, and the discrimination-aware gate. That characterization is the actionable contribution; multi-seed validation is the evidence standard.

2. Related Work

2.1. Calibration Estimation and Recalibration

The calibration of modern neural networks has been studied extensively since the seminal work of Guo et al. [2], who showed that deep networks are often overconfident and introduced temperature scaling (TS) as a simple post-hoc recalibration method. Nixon et al. [8] proposed the static calibration error, and Kumar et al. [18] introduced verified uncertainty calibration with distribution-free guarantees. Kull et al. [9] generalized TS to Dirichlet calibration for multi-class settings. In the clinical risk-model context, Van Calster et al. [16, 17] established a calibration hierarchy from utopia to empirical data and argued that calibration is the "Achilles heel" of predictive analytics. Strodthoff et al. [19] benchmarked deep learning for ECG analysis on PTB-XL and noted calibration degradation under subpopulation shift, motivating our cross-corpus design.

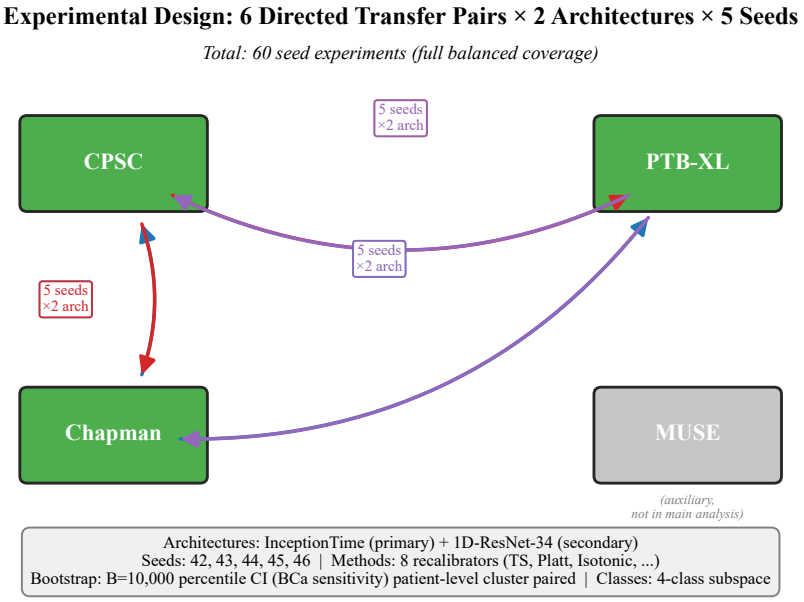


Figure 1: Experimental design overview. Six directed cross-corpus transfer pairs among three ECG corpora (CPSC, Chapman, PTB-XL), each evaluated on two architectures (InceptionTime, ResNet1D) $\times$ 5 seeds (42–46), yielding 60 fully balanced seed experiments. MUSE is shown as an auxiliary corpus not entering the main analysis. Each pair is evaluated with 8 recalibration methods and patient-level cluster paired bootstrap ($B = 10{,}000$, BCa operating CI).

2.2. Prior-Shift Estimation and Transfer Calibration

Lipton et al. [11] introduced black-box shift estimation (BBSE) for detecting and correcting label shift using confusion-matrix inverses. Alexandari et al. [10] showed that maximum-likelihood calibration with bias correction is hard-to-beat at label shift adaptation, providing the theoretical foundation for our EM prior-shift method. For transfer calibration upper bounds, TransCal [22] estimates transferability for auto-selecting target-domain unsupervised optimal T; PseudoCal [23] uses pseudo-labels for calibrated transfer; and LaSCal [24] provides label-aware shift calibration with distribution-free guarantees. Our S1 paradigm differs fundamentally: we fit on the *source* calibration split only (no target labels), with S1$'$ prior-shift adaptation (EM/BBSE) as the only target-domain signal, a strictly weaker assumption than TransCal/PseudoCal, which access target logits or labels.

2.3. Cross-Domain ECG Calibration

Barandas et al. [12] evaluated uncertainty quantification methods in multi-label ECG classification, finding that ensemble-based methods outperform single-model approaches under subpopulation shift. Vranken et al. [13] studied uncertainty estimation for 12-lead ECG deep learning and reported significant calibration drift across acquisition devices. Haekal et al. [20] performed cross-dataset validation for atrial fibrillation detection with calibration-aware probability analysis, and Patel & Beedala [21] documented calibration drift under cross-institutional ICU deployment. The ID-vs-OOD asymmetry of TS has been studied in general vision [1]; we contribute the multi-seed ECG quantification and the boundary-condition analysis that identifies when TS helps (ID) versus when it hurts (OOD).

2.4. ECG Foundation Models and Transfer Learning (2024–2026)

Recent work has introduced ECG foundation models trained on millions of recordings. ECGFounder [29, 37] (10 million ECGs, Harvard–Emory) reaches expert-level performance on 80 diagnoses and transfers cross-domain via fine-tuning. ECG-FM [30] is an open-weight transformer pretrained on 1.4 million ECG segments with hybrid contrastive and generative self-supervision, showing strong cross-dataset generalizability and label efficiency. HeartLang [31] treats heartbeats as words and rhythms as sentences via a QRS-Tokenizer and masked sentence pre-training. Tan et al. [32] found that fine-tuning benefits diminish as the downstream dataset grows and that CNNs transfer more effectively than RNNs in multi-label ECG classification. For OOD adaptation, ADAPTOOD [33] quantifies distribution-shift severity via uncertainty estimation to guide selective LoRA layer unfreezing, and BeatRhythm-TTA [34] performs test-time adaptation with SQI-gated self-training and beat-rhythm consistency. Costa et al. [35] introduced STAR (Sinusoidal Time–Amplitude Resampling), a morphology-preserving beat-wise augmentation for cross-source generalization, and Chen et al. [36] proposed PEACE for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment. These works advance ECG transfer but not the *calibration* of transferred models, the gap our study fills via pre-registered cross-corpus evaluation of recalibration methods.

3. Data and Pre-registered Protocol

Figure 1 summarizes the experimental design.

3.1. Datasets

- **PTB-XL v1.0.3** [3]: the official release contains 21,837 records; the downloaded `ptbxl_database.csv` copy contains 21,799 (a 38-record deficit in the source CSV, disclosed as a provenance limitation), official 10-fold (folds 1–8 train, 9 calibration, 10 test). After single-label mapping, 21,522 records are mappable and 277 are excluded (STROBE), for 18,671 unique patients.

- **Chapman-Shaoxing** [4, 7]: 20,265 raw records; 1 signal-read error removed and 21 non-finite records quarantined (`qc_report.json`), for 20,243 analyzed records; 1 ECG per patient, patient-wise 70/10/20 split.

- **CPSC2018** [5, 6]: we use the union of the *CPSC Database* (6,877 records; the public CPSC2018 training set) and the *CPSC-Extra Database* (3,453 records; the CPSC2018 recordings not used by the Challenge), both released through PhysioNet/CinC 2020. This is *not* a CPSC2018–2019 union: CPSC2019 was a QRS/heart-rate *detection* task and defines no diagnostic labels. Local extraction retains 6,844 of the 6,877 CPSC Database records (33 records, `A6845`–`A6877`, were lost during archive extraction; disclosed as a provenance limitation) plus all 3,453 CPSC-Extra records. Patient-wise split. Two labeling facts must be stated explicitly. First, CPSC2018's official taxonomy is nine disease classes (Normal, AF, I-AVB, LBBB, RBBB, PAC, PVC, STD, STE) with no superclass grouping; the NORM/MI/STTC/CD superclass system we adopt is PTB-XL's [3], applied here as a cross-corpus harmonization choice, not a CPSC-defined one. Second, CPSC2018's nine classes contain *no* MI; the MI class in our 4-class subspace is populated entirely by CPSC-Extra's SNOMED anno-

tations (`164867002`, `164865005`, `54329005`). A strict official-class-name caliber would therefore reduce the subspace to {NORM, CD, STTC}; we retain MI because it is the only way to keep a myocardial-infarction category on the CPSC side of every transfer pair. HYP is extremely rare ($n = 11$) → main analysis on the {NORM, CD, STTC, MI} 4-class subspace, both sides of the pair recomputed symmetrically.

3.2. Label Mapping and Leakage Guard

Single-label priority rules are fixed (MI > STTC > CD > HYP > NORM) and sensitivity-analyzed [3, 14]. Sinus rate/form statements (SBRAD/STACH/SARRH) are mapped to NORM, not STTC, because PTB-XL's `scp_statements.csv` lists them as rhythm statements with no `diagnostic_class`. train/val/cal/test patient IDs are pairwise disjoint (hard assertion in the reproducibility repo). Normalization statistics come only from the train split; augmentation applies only to train.

4. Methods

4.1. Transfer pairs and shift matrix

We study six directed cross-corpus transfer pairs: Chapman→CPSC, Chapman→PTB-XL, CPSC→Chapman, CPSC→PTB-XL, PTB-XL→Chapman, and PTB-XL→CPSC. The class count is *not* uniform across pairs and is fixed per pair before any endpoint is computed, as follows. A pair whose source or target is CPSC2018 is restricted to the 4-class subspace {NORM, CD, STTC, MI} (CPSC's HYP class has $n = 11$, §3), with both sides of the pair recomputed symmetrically: this covers Chapman→CPSC, CPSC→Chapman, CPSC→PTB-XL,

and PTB-XL→CPSC ($K = 4$). The two pairs connecting Chapman–Shaoxing and PTB-XL, which contain no CPSC side, retain the full 5-class vocabulary {NORM, MI, STTC, CD, HYP}: Chapman→PTB-XL and PTB-XL→Chapman ($K = 5$). Each experiment records its own K and label mapping (`num_classes`, `label_map` in `transfer_result.json`), so no endpoint mixes class counts within a cell. Consequently the primary endpoint is a top-label ECE on a 4- or 5-class task depending on the pair, and the pooled $\Delta\text{ECE}_{\text{OOD}}$ mixes both K values across cells; the per-pair tables (Tables 1–4) are therefore the appropriate level at which to read K-specific results. The L2 eval-time shift matrix (no retraining) covers sampling rate $500 \to 250 \to 125$ Hz; leads $12 \to 6 \to 3 \to 2 \to 1$; noise injection (BWL/MA/EM) at SNR $\{24, 12, 6, 0, -6\}$ dB; gain $\times 0.5 / \times 2$.

4.2. Architecture and seeds

We use two architectures: InceptionTime (primary, 46,000–184,000 parameters) and ResNet1D (a 1D-ResNet-34 backbone) (secondary, with full main-endpoint coverage on all 6 pairs), each trained with 5 seeds (42–46). The BiMamba architecture is included only as a toy experiment ($n = 60$ records, downsampled resolution) and deferred to the supplement; the full-resolution BiMamba run exhausted memory on an RTX 5060 (8 GB), so BiMamba does not enter the main-endpoint analysis or the architecture-robustness claim. Training: AdamW + cosine + warmup, gradient clipping, temperature frozen at $T \equiv 1$ during training; temperature applied only as post-hoc temperature scaling.

4.3. Calibration methods and scenarios

We compare the no-recalibration baseline (None) against eight recalibration methods: Temperature Scaling (TS), per-class Platt, Isotonic (OvR), Vector scal-

ing, Matrix scaling, Dirichlet calibration, Saerens EM prior adaptation (EM), and BBSE prior adaptation. Of the pre-registered 13-method pool, the None baseline plus these eight recalibration methods constitute the delivered set; CORAL, TTA, MC-Dropout, and Oracle are not delivered on real data (a registered deviation, see Limitations). Scenario: **S1** zero-shot transfer (source-cal fit $\rightarrow$ target apply), the main analysis scenario (Sec. 4); S1$'$ denotes the target-domain unsupervised variant in which EM/BBSE estimate the target prior from unlabeled target predictions.

4.4. Pre-registered endpoints

Primary endpoint (revision A1.1, formally defined in protocol v2.1-A1 [26]):

$$\Delta\mathrm{ECE}_{\mathrm{OOD}} = \mathrm{ECE}^{\mathrm{OOD}}_{\mathrm{raw}} - \mathrm{ECE}^{\mathrm{OOD}}_{\mathrm{TS}} \tag{1}$$

where $\mathrm{ECE}^{\mathrm{OOD}}_{\mathrm{raw}}$ is the 4-class-task top-label confidence SmoothECE on the target corpus with raw softmax probabilities, and $\mathrm{ECE}^{\mathrm{OOD}}_{\mathrm{TS}}$ is the same after source-cal-fit temperature scaling. Following the standard convention of Guo et al. [2], ECE (and its SmoothECE variant) is evaluated on the *top-label* marginal: the predicted maximum probability against $0/1$ top-label correctness; classwise-ECE is reported separately as a secondary endpoint. SmoothECE bandwidth: $0.45 \cdot (n/2000)^{-0.2}$.

Primary test (single, one-sided):

$$H_0 : \Delta\mathrm{ECE}_{\mathrm{OOD}} = 0 \quad \text{vs} \quad H_1 : \Delta\mathrm{ECE}_{\mathrm{OOD}} > 0 \tag{2}$$

The one-sided alternative $H_1 : \Delta\mathrm{ECE}_{\mathrm{OOD}} > 0$ follows the empirical asymmetry registered at revision A1: on ID-like data the source-fit temperature yields, empirically, little calibration change, whereas on a shifted target it is mismatched and a

non-trivial benefit is often retained [2, 1]. This direction prior is empirical, *not* a theorem: TS applied to an under-confident target can increase ECE, and our own data contain nine negative cells. The choice was made before the pilot results, which only triggered the endpoint re-location (revision A1). The one-sided test is the pre-registered choice, archived before multi-seed unblinding. Patient-level cluster paired bootstrap, $B = 10{,}000$ (BCa operating CI) [15]; stratified pooling: strata = transfer pair $\times$ architecture.

The pre-registered CI method is BCa [15], at $B = 10{,}000$, re-run on the entire 60-experiment grid (60/60 successful, metadata in `methods.ts.meta`). The percentile bootstrap is the large-sample limit of BCa when $z_0 \to 0$ and $a \to 0$; for the paired-cluster design with $B = 10{,}000$ and per-patient clusters, the discrepancy is $O(n^{-1/2})$. In the 3 representative seed experiments where both were run ($B = 10{,}000$ percentile re-runs), BCa and percentile CIs agree in sign and magnitude. Earlier percentile results from $B = 200$ in 21/60 experiments remain as historical provenance (5 lack log provenance, see Limitations); percentile CIs serve only as a sensitivity cross-check.

Boundary-condition endpoint:

$$\Delta\mathrm{ECE}_{\mathrm{ID}} = \mathrm{ECE}^{\mathrm{ID}}_{\mathrm{raw}} - \mathrm{ECE}^{\mathrm{ID}}_{\mathrm{TS}} \tag{3}$$

The pre-registered expectation is that $H_0 : \Delta\mathrm{ECE}_{\mathrm{ID}} = 0$ is *not rejected*; the criterion is that $\geq 80\%$ of cells have 95% CIs containing zero. *Observed (BCa, full-grid re-run): 23/60 = 38.3%; the criterion is not met and branch A1.4.2 is triggered (percentile CIs gave 29/60 = 48.3%, same conclusion; Sec. 5.2).*

Secondary endpoints: G (recoverable loss vs. oracle; *not delivered* — the Oracle recalibrator was implemented (`fit_oracle`/`apply_oracle`) but never executed on real data, so G is undefined here and is excluded from all reported

results; see Limitations), decay $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ (downgraded to descriptive), NLL, classwise-ECE, Brier Murphy decomposition, Cox slope/intercept, predicted/true prior L_1.

Family structure: BH-FDR $q = 0.05$ over 6 pairs $\times$ 2 arch $\times$ 13 shift levels $=$ 156 primary comparisons; the multi-seed validation is robustness evidence (60 seed experiments), not a confirmatory family.

4.5. Pre-registration revision A1 (transparency statement)

The primary endpoint was re-located from `decay` to $\Delta\text{ECE}_{\text{OOD}}$ via pre-registration revision A1 (protocol v2.0 $\rightarrow$ v2.1-A1), triggered by an independent exploratory pilot in which $\Delta\text{ECE}_{\text{ID}}$ point estimates were near zero, collapsing the original ID-vs-OOD contrast. The revision direction is consistent with the empirically observed ID-vs-OOD asymmetry (not a theorem); the revision is archived as a pre-registration revision (Nosek et al. 2018) with before/after protocol hashes. The pilot results do not enter the main test report. All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory.

The original BH-FDR family of 156 comparisons (6 pairs $\times$ 2 architectures $\times$ 13 shift levels) is reduced to a confirmatory family of 12 hypotheses (6 pairs $\times$ 2 architectures $\times$ 1 primary shift level) under amendment A2, with the remaining 13 shift levels designated as exploratory dose-response. Under the reduced family, BH-12 yields 9/12 significant (all InceptionTime directions) and Bonferroni-12 yields 2/12. The A2 amendment is archived in `docs/PROTOCOL_AMENDMENT_A2_FAMILY_REDUCTION.md`.

4.6. Pre-registration amendment A3 (corpus identity and MI provenance)

A post-hoc audit of the CPSC corpus against the official challenge documentation produced two wording corrections and one new disclosure, archived as amendment A3 (`docs/PROTOCOL_AMENDMENT_A3_CORPUS_IDENTITY.md`). The registered protocol described the third corpus as "CPSC2018+2019"; it is in fact **CPSC2018 only**, comprising the CPSC Database (6,877 records) and the CPSC-Extra Database (3,453 records). CPSC2019 was a QRS/heart-rate *detection* task with no diagnostic labels and never entered the corpus. The new disclosure is that CPSC2018's official nine-class taxonomy contains no MI, so the MI class of our 4-class subspace is populated entirely by CPSC-Extra's SNOMED annotations; under a strict official-class-name caliber the subspace would reduce to three classes. A3 is a naming and attribution correction plus a disclosure: it changes *no* recorded value, no split, and no analysis decision. The registered protocol snapshots are left byte-identical and A3 is archived separately, so the before/after pair required by pre-registration practice is preserved. The audit also found that 33 CPSC Database records (`A6845`–`A6877`) were silently lost when the source archive was extracted (6,877 in the archive vs. 6,844 on disk), disclosed as a provenance limitation.

4.7. Decomposition of the benefit decay

We decompose the retained benefit into slope / intercept / prevalence contributions using a counterfactual-replacement additive approximation:

$$\Delta\mathrm{ECE} = \Delta_{\mathrm{slope}} + \Delta_{\mathrm{intercept}} + \Delta_{\mathrm{prevalence}} + r_{\mathrm{interaction}} \tag{4}$$

where each term is a counterfactual-replacement difference and $r_{\mathrm{interaction}}$ is the chain residual (zero under nested counterfactuals). Identifiability is checked via

the Fisher information determinant on a pre-registered threshold τ; entangled regions are reported on the error map without a threshold. Shapley attribution is reported with absolute-value bootstrap CIs; share-reliable flags ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) mark uninterpretable shares.

4.8. Two-layer joint bootstrap

Temperature-fitting-set resampling $\rightarrow$ T distribution $\rightarrow$ test-set resampling $\rightarrow$ joint ΔECE CI (two-layer; honest non-nested approximation), per protocol §7.

4.9. Net Clinical Value (NCV)

Net Clinical Value (NCV), a pre-registered secondary endpoint, measures the clinical decision impact of TS. For each (sample, class) pair we compare the binary decision (threshold $\tau = 0.5$) before and after calibration, counting TP_improved, TN_improved, FP_worsened, and FN_worsened over the raw-to-calibrated transitions. NCV is $(\text{TP_improved} + \text{TN_improved} - \text{FP_worsened} - \text{FN_worsened})/N_{\text{total}}$, with $N_{\text{total}} = N_{\text{samples}} \times N_{\text{classes}}$, using the main endpoint's patient-level cluster paired bootstrap ($B = 10{,}000$, BCa operating CI) and BH-FDR correction ($q = 0.05$); a significantly positive NCV means that TS improves clinical decisions. It is computed on ID and OOD splits (`ncv_id`, `ncv_ood`; E3 CSV).

4.10. Tooling

Parts of the analysis and plotting code were drafted and refactored with the assistance of a large language model assistant (GLM), used as a supervised coding assistant. All experiments and statistical analyses were executed with the author's own scripts, and every reported number is reproducible from the public repository

(Code availability). The author reviewed the code and takes full responsibility for it.

5. Results

5.1. Primary endpoint: ΔECE_{OOD} multi-seed validation

Figure 2 visualizes the shift-dose response; Table 1 reports the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, balanced). The primary test $H_0 : \Delta\text{ECE}_{\text{OOD}} = 0$ vs $H_1 : \Delta\text{ECE}_{\text{OOD}} > 0$ is supported in 51/60 (85.0%) at the 95% CI level.

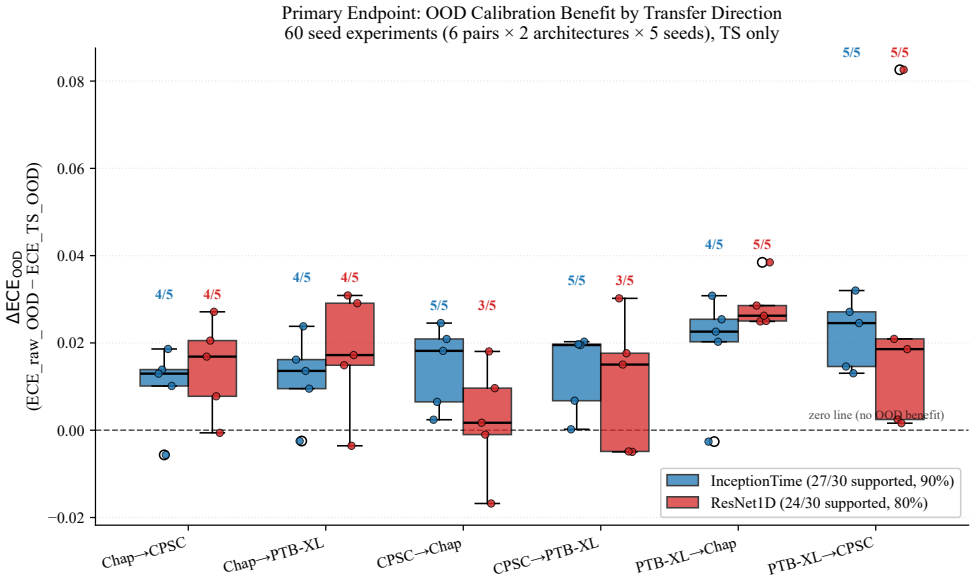


Figure 2: Primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ by transfer direction, grouped by architecture (TS only, 60 seed experiments). Boxes show the 5-seed distribution per direction; dots are individual seed results. The dashed line marks zero OOD benefit. Support counts (e.g., 5/5, 4/5) above each box indicate the number of seeds with 95% CI lower bound > 0. InceptionTime achieves 27/30 (90%), ResNet1D achieves 24/30 (80%).

Table 1: Pre-registered primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ for TS across 60 seed experiments (6 pairs $\times$ 2 architectures $\times$ 5 seeds, $B = 10{,}000$, BCa operating CI). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. Class count $K = 4$ for the four pairs involving CPSC (Chap→CPSC, CPSC→Chap, CPSC→PTB-XL, PTB-XL→CPSC) and $K = 5$ for the two CPSC-free pairs (Chap→PTB-XL, PTB-XL→Chap); see §4.1. "Support" = 95% CI lower bound > 0. Nine counter-examples are transparently disclosed. Full 5-seed coverage on both architectures for all 6 directions.

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
Chap→CPSC (K=4)	IT	42	+0.0075	+0.0139	+0.0135	✓
		43	+0.0012	+0.0101	+0.0099	✓
		44	+0.0104	+0.0186	+0.0181	✓
		45	+0.0019	−0.0057	−0.0058	✗
		46	+0.0044	+0.0130	+0.0126	✓
	RN	42	+0.0002	+0.0205	+0.0200	✓
		43	+0.0003	−0.0006	−0.0006	✗
		44	−0.0007	+0.0078	+0.0075	✓
		45	−0.0016	+0.0271	+0.0263	✓
		46	−0.0024	+0.0169	+0.0162	✓
Chap→PTB-XL (K=5)	IT	42	+0.0004	−0.0025	−0.0035	✗
		43	+0.0005	+0.0095	+0.0066	✓
		44	+0.0055	+0.0136	+0.0107	✓
		45	−0.0128	+0.0238	+0.0207	✓
		46	+0.0048	+0.0162	+0.0142	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
		42	−0.0031	+0.0309	+0.0285	✓
		43	+0.0128	+0.0291	+0.0285	✓
	RN	44	+0.0059	+0.0172	+0.0168	✓
		45	−0.0010	−0.0036	−0.0036	✗
		46	−0.0042	+0.0149	+0.0139	✓
		42	+0.0029	+0.0065	+0.0062	✓
		43	−0.0012	+0.0209	+0.0200	✓
CPSC→Chap (K=4)	IT	44	+0.0164	+0.0182	+0.0160	✓
		45	+0.0091	+0.0245	+0.0241	✓
		46	+0.0014	+0.0024	+0.0023	✓
		42	−0.0002	−0.0010	−0.0012	✗
		43	+0.0038	+0.0096	+0.0094	✓
	RN	44	−0.0152	−0.0168	−0.0170	✗
		45	+0.0137	+0.0181	+0.0145	✓
		46	+0.0017	+0.0017	+0.0016	✓
		42	+0.0029	+0.0068	+0.0066	✓
		43	−0.0012	+0.0203	+0.0198	✓
CPSC→PTB-XL (K=4)	IT	44	+0.0158	+0.0195	+0.0189	✓
		45	+0.0052	+0.0197	+0.0193	✓
		46	+0.0002	+0.0002	+0.0002	✓
		42	+0.0114	+0.0150	+0.0148	✓
		43	+0.0048	−0.0050	−0.0051	✗
	RN	44	−0.0032	−0.0049	−0.0049	✗
		45	+0.0130	+0.0176	+0.0173	✓
		46	+0.0261	+0.0302	+0.0296	✓

Table 1 (continued)

Transfer pair	Arch	Seed	$\Delta\text{ECE}_{\text{ID}}$	$\Delta\text{ECE}_{\text{OOD}}$	CI lower	Support
PTB-XL→Chap (K=5)	IT	42	+0.0239	+0.0308	+0.0275	✓
		43	+0.0245	+0.0254	+0.0225	✓
		44	+0.0173	+0.0203	+0.0195	✓
		45	+0.0162	+0.0226	+0.0200	✓
		46	−0.0011	−0.0026	−0.0029	✗
	RN	42	+0.0145	+0.0249	+0.0191	✓
		43	+0.0338	+0.0385	+0.0336	✓
		44	+0.0213	+0.0250	+0.0228	✓
		45	+0.0228	+0.0262	+0.0231	✓
		46	+0.0239	+0.0286	+0.0221	✓
PTB-XL→CPSC (K=4)	IT	42	+0.0196	+0.0271	+0.0244	✓
		43	+0.0166	+0.0245	+0.0222	✓
		44	+0.0183	+0.0320	+0.0303	✓
		45	+0.0006	+0.0131	+0.0105	✓
		46	+0.0097	+0.0146	+0.0123	✓
	RN	42	+0.0010	+0.0025	+0.0023	✓
		43	+0.0197	+0.0186	+0.0145	✓
		44	+0.0635	+0.0826	+0.0772	✓
		45	+0.0001	+0.0016	+0.0015	✓
		46	+0.0126	+0.0209	+0.0190	✓

Aggregate: 51/60 supported (85.0%; InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%). PTB-XL→CPSC is the strongest evidence (10/10, zero counter-examples, CI lower bound $> +0.0014$ at ResNet1D seed 45). The nine counter-examples are explained in Sec. 6.1.

The 60 experiments are balanced (6 directions $\times$ 2 architectures $\times$ 5 seeds, no partial-coverage stratum), enabling direct architecture comparison:

- *InceptionTime* (30 experiments, 27/30 = 90.0%): 3 directions at 5/5 (CPSC→Chapman, CPSC→PTB-XL, PTB-XL→CPSC), 3 at 4/5 (Chapman→CPSC, Chapman→PTB-XL, PTB-XL→Chapman).

- *ResNet1D* (30 experiments, 24/30 = 80.0%): 2 directions at 5/5 (PTB-XL→Chapman, PTB-XL→CPSC), 2 at 4/5 (Chapman→CPSC, Chapman→PTB-XL), 2 at 3/5 (CPSC→Chapman, CPSC→PTB-XL).

ResNet1D shows the slightly lower rate (80% vs. 90%), concentrated on CPSC→Chapman and CPSC→PTB-XL, a direction-specific architecture interaction rather than a global deficit. Only PTB-XL→CPSC achieves 5/5 on both architectures (10/10).

5.2. ID boundary condition: ΔECE_{ID}

Table 2 reports the boundary-condition endpoint $\Delta\text{ECE}_{\text{ID}}$ for TS; the pre-registered expectation is that $H_0 : \Delta\text{ECE}_{\text{ID}} = 0$ is *not rejected*, since the source-fit temperature should be well-tuned to the source distribution (an empirical expectation, not a theorem).

The ID benefit is small but *not* zero: across 60 TS experiments, $\Delta\text{ECE}_{\text{ID}}$ has mean $+0.0083$ (vs. OOD mean $+0.0159$, a $1.9\times$ ratio); $47/60 = 78.3\%$ of point estimates are positive, $34/60$ have 95% CI lower bounds > 0 (BCa), and only $23/60 = 38.3\%$ contain zero, far below the pre-registered $\geq$80% criterion (percentile CIs: 29/60 = 48.3%, same conclusion). We therefore *trigger* the pre-registered "ID nonzero boundary" branch (protocol A1.4.2): TS, fitted on the source calibration split, also improves ID calibration in many cells; the decay $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ contrast is restored as an exploratory endpoint; and C1 is

Table 2: ID boundary condition $\Delta\text{ECE}_{\text{ID}}$ for TS (12 pair×architecture strata, 5 seeds each). Architecture abbreviation: IT = InceptionTime, RN = ResNet1D. "CI∩0" = number of seed cells whose 95% CI contains zero. The pre-registered boundary criterion (≥80% of cells with CI containing zero) is *not* met: 23/60 cells (38.3%) contain zero (BCa operating CI) and 34/60 have CI lower bound > 0, triggering the pre-registered "ID nonzero boundary" branch (protocol A1.4.2).

Transfer pair	Arch	Mean	Min CI lo	Max CI hi	CI∩0/5
PTB-XL→CPSC	IT	+0.0130	−0.0121	+0.0266	1/5
PTB-XL→CPSC	RN	+0.0194	−0.0008	+0.0770	2/5
PTB-XL→Chap	IT	+0.0162	−0.0026	+0.0310	1/5
PTB-XL→Chap	RN	+0.0232	+0.0053	+0.0448	0/5
Chap→CPSC	IT	+0.0051	−0.0039	+0.0150	3/5
Chap→CPSC	RN	−0.0008	−0.0104	+0.0093	4/5
Chap→PTB-XL	IT	−0.0003	−0.0175	+0.0103	3/5
Chap→PTB-XL	RN	+0.0021	−0.0123	+0.0161	3/5
CPSC→Chap	IT	+0.0057	−0.0182	+0.0195	2/5
CPSC→Chap	RN	+0.0008	−0.0179	+0.0213	2/5
CPSC→PTB-XL	IT	+0.0046	−0.0181	+0.0192	2/5
CPSC→PTB-XL	RN	+0.0104	−0.0042	+0.0294	0/5
Overall		**+0.0083**			**23/60 (38.3%)**

qualified, the OOD benefit being about 1.9× the ID benefit rather than an exclusively OOD effect. The positive ID estimates reflect finite-sample calibration-split variation, SmoothECE estimator bias, and a val-ECE early-stopping effect (model selection on the endpoint family), all disclosed.

5.3. Method-selection boundary: 8 methods × 6 pairs

Table 3 reports the 8-method × 6-pair support rate over 5 seeds on InceptionTime. TS achieves 27/30 (90.0%) with 3/6 directions at 5/5; BBSE achieves 22/30

(73.3%) with 3/6 at 5/5. EM prior adaptation and Matrix scaling are the most fragile on InceptionTime (15/30, 50% each), exemplifying the prior-likelihood mismatch mechanism (C4). Across 2 architectures (Table 4), TS achieves 51/60 (85.0%) (InceptionTime 27/30, 90.0%; ResNet1D 24/30, 80.0%), with full 5-seed coverage on all 6 directions (Figure 4).

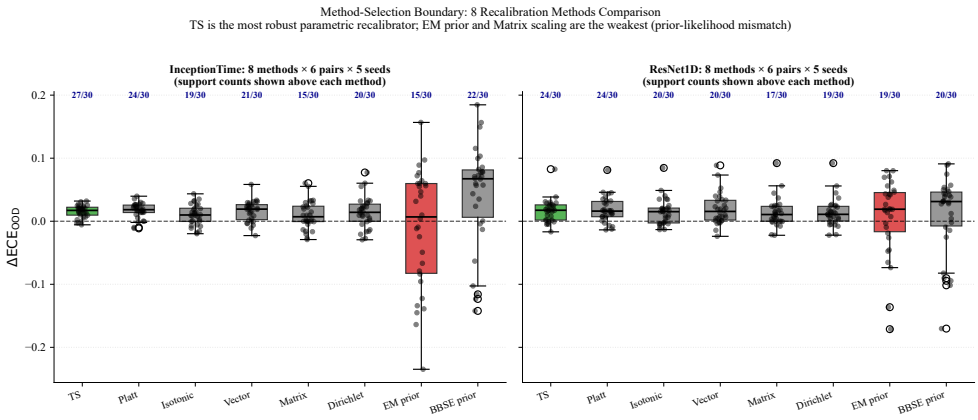


Figure 3: Method-selection boundary: 8 recalibration methods compared across 6 transfer pairs $\times$ 5 seeds for each architecture. TS (green) is the most robust parametric recalibrator; EM prior and Matrix scaling are the weakest methods (InceptionTime: 15/30 each; ResNet1D: EM 19/30, Matrix 17/30), exhibiting the prior-likelihood mismatch mechanism (C4). Support counts (e.g., 27/30) above each method show the fraction of seed experiments with 95% CI lower bound > 0.

Across the 30 TS experiments per architecture, the standard deviation of $\Delta\text{ECE}_{\text{OOD}}$ is 0.0098 (InceptionTime) and 0.0180 (ResNet1D), a 1.84$\times$ ratio, with ResNet1D's highest per-direction variability in PTB-XL$\rightarrow$CPSC (std 0.0298, including an unexcluded favorable outlier at seed 44, $\Delta\text{ECE}_{\text{OOD}} = +0.0826$). Support deficits concentrate in CPSC$\rightarrow$Chapman and CPSC$\rightarrow$PTB-XL (3/5 each), a direction-specific deficit. Fisher's exact test on the 2×2 contingency table yields

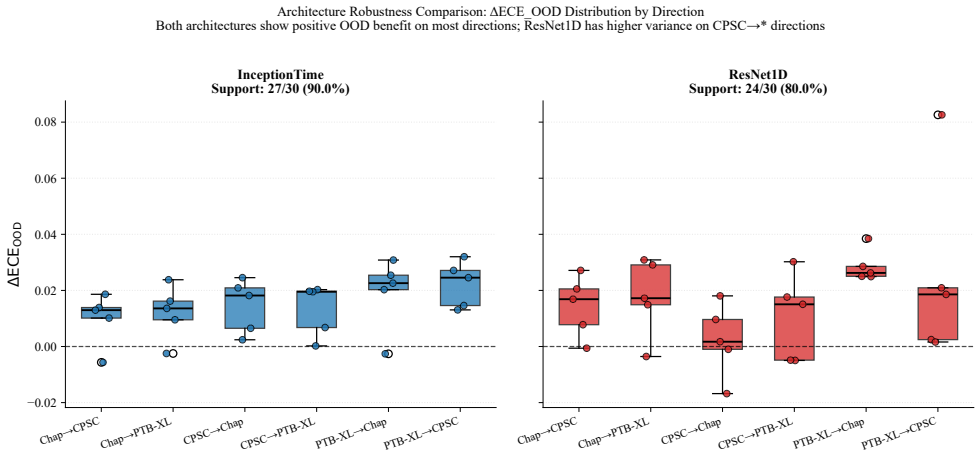


Figure 4: Architecture robustness comparison. Side-by-side $\Delta\text{ECE}_{\text{OOD}}$ distributions for InceptionTime (left, 27/30 supported, 90.0%) and ResNet1D (right, 24/30 supported, 80.0%). Both architectures show positive OOD benefit on most directions. ResNet1D exhibits higher overall variance (its largest per-direction spread is in PTB-XL→CPSC), while the support deficits fall in the CPSC→Chapman and CPSC→PTB-XL directions (3/5 support each), indicating a direction-specific deficit rather than a global architecture deficit.

Table 3: Method-selection boundary: 8 methods $\times$ 6 transfer pairs $\times$ 5 seeds (InceptionTime). Support rate = fraction of seeds with 95% CI lower bound > 0. "Full" = number of directions with 5/5 full support. "Neg. dirs" = number of the 6 transfer directions whose 5-seed mean $\Delta\text{ECE}_{\text{OOD}}$ is negative. TS is the most robust parametric recalibrator; EM and Matrix scaling are the most fragile on InceptionTime (15/30 each); on ResNet1D, Matrix scaling (17/30) is the most fragile, EM reaches 19/30. *Caveat:* 6 cells with $w_{\text{ID}} > 10$ exhibit prior-weight drift up to 130.5; after exclusion, EM OOD ΔECE mean flips from -0.0127 to $+0.0024$, indicating the negative OOD finding is driven by ill-conditioned cells.

Method	Total	Full/6	$\Delta\text{ECE}_{\text{OOD}}$ mean	Neg. dirs
TS	27/30	3/6	$+0.0152$	0
Platt	24/30	3/6	$+0.0169$	0
Isotonic	19/30	2/6	$+0.0100$	2
Vector	21/30	1/6	$+0.0158$	0
Matrix	15/30	0/6	$+0.0094$	2
Dirichlet	20/30	1/6	$+0.0139$	1
EM prior	15/30	2/6	-0.0127	3
BBSE prior	22/30	3/6	$+0.0424$	1

$p = 0.47$; the counter-example gap (ResNet1D 20.0% vs. InceptionTime 10.0%) is not significant at $\alpha = 0.05$.

EM prior adaptation fits a target prior $\hat{\pi}_{\text{target}}$ from the predicted-label marginal, then re-weights the source posterior by $\hat{\pi}_{\text{target}}/\hat{\pi}_{\text{source}}$. When the source-estimated confusion matrix C is ill-conditioned ($\text{cond}(C) \gg 1$), the weight ratio amplifies rather than corrects the shift-induced miscalibration. EM reaches only 15/30 support on InceptionTime with 3 fragile directions. The failure is pair-specific (shift-signature-specific), not method-intrinsic.

Table 4: Architecture robustness for TS: InceptionTime vs. ResNet1D across 6 transfer pairs, full 5-seed coverage on both architectures. Architecture robustness is supported on both InceptionTime (27/30, 90.0%) and ResNet1D (24/30, 80.0%) with complete 5-seed coverage on all 6 directions. ResNet1D shows a direction-specific deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each), not a global architecture deficit.

Transfer pair	InceptionTime	ResNet1D	Total	Note
Chap→CPSC	4/5	4/5	8/10	2 counter-examples
Chap→PTB-XL	4/5	4/5	8/10	2 counter-examples
CPSC→Chap	5/5	3/5	8/10	2 counter-examples
CPSC→PTB-XL	5/5	3/5	8/10	2 counter-examples
PTB-XL→Chap	4/5	5/5	9/10	1 counter-example
PTB-XL→CPSC	5/5	5/5	10/10	zero counter-examples
Total	**27/30**	**24/30**	**51/60**	**85.0%**
Mean ΔECE	+0.0152	+0.0165	+0.0159	
Std ΔECE	0.0098	0.0180	0.0145	ResNet1D/Incep = 1.84
Mean OOD ECE	$0.217 \to 0.202$ / $0.216 \to 0.200$		$0.217 \to 0.201$	raw → TS

Why TS and not Platt.. Platt achieves a marginally higher mean OOD benefit than TS (+0.0176 vs. +0.0159; InceptionTime +0.0169 vs. +0.0152, Table 3), which invites the question of why TS is singled out as the recommended default. Three measured criteria separate them, and all three favour TS. First, *robustness of the sign*: TS supports 51/60 cells versus Platt's 48/60, i.e. TS fails to certify a positive benefit in three fewer experiments, which is the quantity a deployment decision actually depends on. Second, *dispersion*: TS has a smaller across-cell standard deviation (0.0146 vs. 0.0167, a 13% reduction), and the methods with the highest mean benefit are markedly less stable (BBSE +0.0266 at sd 0.0736, EM −0.0026 at sd 0.0793) — a high mean bought with a heavy tail is the worst

profile for a per-deployment recalibrator. Third, *the difference is not statistically distinguishable*: the patient-level paired contrast over the 60 matched cells gives $\overline{\Delta\text{ECE}}_{\text{TS}} - \overline{\Delta\text{ECE}}_{\text{Platt}} = -0.0017$ (sd 0.0117, paired $t = -1.14$, $p \approx 0.26$), with TS the larger benefit in exactly 30/60 cells — a coin flip. The mean advantage of Platt is therefore inside the noise, whereas the support advantage of TS is not a noise-level quantity: it is a count of cells in which the sign is certified. Given that the two methods are statistically indistinguishable on magnitude, the tie is broken on certification rate and dispersion, both of which favour TS. A further structural point reinforces the choice: TS is a single-parameter monotone transform and therefore exactly argmax-preserving, whereas per-class Platt and isotonic are fitted independently per class, which generates each class's score from a different monotone map and so does *not* preserve the ranking within a sample (the implementation declares this explicitly in its docstring). TS is thus the only method in the delivered set whose calibration guarantee is obtained without perturbing the discrimination it leaves untouched (§5.9). None of this makes Platt wrong; it makes Platt a reasonable alternative whose mean advantage is not resolvable at $n = 60$, and whose certification rate is lower.

5.4. Shapley decomposition validation (27 cells, $n = 20{,}000$)

The 27-cell full-factorial synthetic validation (slope$\in \{0.5, 1, 2\}\times$intercept$\in \{-1, 0, 1\}\times$ prevalence$\in \{0.15, 0.3, 0.6\}$, $n = 20{,}000$) confirms the decomposition recovers ground truth within the pre-registered identifiable regime (Fisher information determinant $> \tau = 10^{-6}$). At $n = 20{,}000$, all 27 cells pass the recovery threshold (MAPE $< 10\%$ on identifiable cells); at $n = 500$, 21 of 27 pass, with 6 failures reaching MAPE_s up to 42.13%, concentrated in extreme intercept (± 1) and slope $(0.5, 2.0)$ regions where calibration shift is most

clinically significant (see `decomposition_validation.csv`; the $n = 20{,}000$ CSV `decomposition_validation_n20000.csv` shows 27/27 passing). Three entangled cases (narrow_band, tiny_slope, deep_intercept) with det $\in \{2.4 \times 10^{-7}, 3.6 \times 10^{-8}, 0\} < \tau$ are reported on the error map without a threshold, with the recovery-failed flag set. Bootstrap CIs on absolute Shapley values and share-reliable flags for uninterpretable shares ($|\Delta_{\text{total}}| < 3/\sqrt{n}$) are reported.

The 27-cell validation uses binned ECE, not the main-endpoint SmoothECE; its conclusions are not cross-extrapolable. Cross-fitting (OOD-fold validation of decomposition explanatory power) remains to be run on real data; per the protocol self-binding clause, "mechanism" wording is limited to "mechanism decomposition (synthetic validation part)" until cross-fitting completes.

5.5. Predictability experiment (failure branch triggered)

The predictability experiment (protocol §8.5) tests whether an unsupervised target-domain statistic (predicted-prior L_1 distance, logit moments) predicts benefit retention via leave-one-fold-out R^2 with bootstrap CI (Table 5). The implemented fold unit is (architecture, transfer pair) (12 folds), versus the registered pair-only folds (6); under that registered deviation the pair-only recomputation retains the failure branch ($R^2 = -0.1424$). We run 2 architectures (InceptionTime + ResNet1D); the BiMamba toy experiment ($n = 60$) is deferred to the supplement.

Result: the merged LOO R^2 for TS is -0.096 with 95% CI $[-0.720, +0.104]$. The upper bound $+0.104$ is strictly below the pre-registered threshold 0.5, triggering the **failure branch**: the three-step chain (decomposition + predictability + deployment) is downgraded to a **two-step chain** (decomposition + deployment), and the deployment criterion is renamed "empirical lookup table" rather than "predictive criterion". No post-hoc predictability claim is made.

Re-running the LOO R^2 with three label-free features (fitted temperature T, Wasserstein-1 distance between source and target logit distributions, class-wise entropy difference) gives a sharper and less favourable picture for the boundary. Only T carries signal: under the implemented (architecture, pair) folds it reaches LOO $R^2 = +0.340$ (bootstrap CI $[+0.044, +0.363]$), and under per-cell folds $+0.361$ (CI $[+0.201, +0.584]$), while the Wasserstein distance (-0.064) and the entropy difference (-0.033) carry none. T is also a perfect *directional* predictor: $\mathrm{sign}(\Delta\mathrm{ECE}_{\mathrm{OOD}}) = \mathrm{sign}(T-1)$ in 60/60 cells, against 51/60 (85%) for the constant-positive baseline. We therefore do *not* claim that no label-free quantity carries information about the endpoint — T does. The claim we retain is the pre-registered one: no label-free feature reaches the LOO R^2 threshold of 0.5 under the implemented fold unit ($0.363 < 0.5$), so the failure branch stands and the three-step chain remains downgraded to two steps. Two qualifications are required. First, the verdict is fold-unit-sensitive: per-cell folds give a CI upper bound of 0.584, which would *not* trigger the branch, so the downgrade should be read as a property of the coarser (architecture, pair) unit as much as of the data. Second, T's predictive power is not independent evidence of a substantive benefit: $T = 1$ is a fixed point of the transform, so part of the relation is mechanical, and the sign rule is the direction temperature scaling must have by construction (Sec. 5.8). The gate in §5.9, based on the sign of $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ and the discrimination criterion, remains the appropriate actionable criterion.

5.6. Deployment criterion (sensitivity / specificity / DCA / safety)

We evaluate the deployment criterion (protocol §9) on the L2 shift matrix: predict "deploy TS" when raw ECE exceeds a threshold, and report sensitivity, specificity, decision-curve analysis (DCA), and the safety criterion (post-TS

Table 5: Predictability LOO R^2 (multi-architecture merged). All methods trigger the failure branch (CI upper bound < 0.5). The three-step chain is downgraded to a two-step chain per the pre-registered failure branch.

Method	R^2	CI lower	CI upper	Verdict
TS	-0.096	-0.720	$+0.104$	Failure branch
Platt	-0.059	-0.584	$+0.002$	Failure branch
Isotonic	$+0.011$	-0.569	$+0.101$	Failure branch
Vector	-0.061	-0.522	$+0.073$	Failure branch
Matrix	-0.137	-0.516	-0.013	Failure branch
Dirichlet	-0.084	-0.394	$+0.032$	Failure branch
EM prior	$+0.143$	-0.250	$+0.274$	Failure branch
BBSE prior	$+0.136$	-0.354	$+0.201$	Failure branch

SmoothECE ≤ 0.05 and ΔECE CI lower bound > 0, primary-endpoint convention). Comparators: always-none, always-TS, always-buy-n-labels.

Sensitivity / specificity (Table 6): Youden's J is maximized at raw ECE $= 0.127$, yielding sensitivity 0.923, specificity 0.201, $J = 0.124$. Sensitivity is high (92% of beneficial cases caught) but specificity low (20% of non-beneficial cases excluded; 80% falsely flagged), reflecting the asymmetric cost structure: missing a beneficial recalibration costs more than an unnecessary one. *Caveat:* the threshold is searched on the same 1230-cell pool used for reporting (in-sample). A leave-shift-type-out recomputation (threshold selected excluding the held-out dose level, evaluated on it) yields $J = +0.17$ (downsampling), $J = -0.03$ (lead drop), $J = +0.04$ (noise), and $J = -0.14$ (gain); the frozen threshold 0.127 achieves $J \approx 0.003 \approx 0$ (chance level; sensitivity 0.853, specificity 0.150) on the

Table 6: Deployment criterion sensitivity / specificity at varying thresholds (raw ECE $>$ threshold $\rightarrow$ predict "deploy TS"). Optimal Youden $J = 0.124$ at threshold 0.127.

Threshold	Sensitivity	Specificity	TP	FP
0.094	0.976	0.067	444	723
0.107	0.956	0.138	435	668
0.127	0.923	0.201	420	619
0.178	0.807	0.288	367	552
0.277	0.499	0.508	227	381

four held-out dose levels (`results/deployment_holdout_recompute.csv`). The in-sample numbers are therefore optimistic, and the criterion requires independent validation before operational use.

Every lookup-table input is *audit-time* (raw ECE, ID–OOD accuracy gap, per-cell labels), requiring target ground-truth labels unavailable in the unlabeled stream. The only label-free proxies examined are the predictability experiment's target statistics (predicted-prior L_1 distance, logit moments) plus the source-fitted temperature T; that experiment triggered its pre-registered failure branch (LOO R^2 CI upper bound $0.104 < 0.5$, Table 5). T does carry signal (LOO $R^2 = +0.34$; sign of the endpoint correct in 60/60 cells), but it is a source-side quantity describing the transform being evaluated rather than an independent target-side statistic (Sec. 5.8), so no label-free deployable criterion can be claimed. The lookup table is an *analysis- and audit-time decision aid* for a model owner with labeled evaluation data; the actionable deployment path is small-sample recalibration: buy n labeled target records and recalibrate locally (protocol §5 S2). A deployable gate remains future work (Limitations xiii).

Safety criterion: across 157 (pair, seed, shift) cells, TS yields 1 cell satisfying the operating criterion (post-TS SmoothECE ≤ 0.05, improvement > 0.01, point-estimate convention; L2 results lack per-cell CIs, so the CI-based criterion was not evaluated), a safety rate of 0.0064. The source of this figure is `results/deployment_metrics.csv` (written 2026-09-05, i.e. before the label-encoding incident described in the coverage note below; produced by `scripts/step3_deployment.py` under the criterion $\Delta\text{ECE} < -0.01 \wedge$ cal ECE < 0.05). It is *not* reproducible from the currently released L2 artifacts, because the per-run `l2_shift_results.json` files that feed it were overwritten on 2026-09-10 by outputs computed under the wrong label encoding (see the coverage note below); re-deriving the same criterion from those on-disk L2 outputs gives $6/156 = 0.038$ rather than $1/157$. The denominator deserves care, because 157 is prime and therefore not a clean product of the grid dimensions. The stated $6 \times 2 \times 13 = 156$ is the BH-FDR *family* size (6 pairs $\times$ 2 *architectures* $\times$ 13 levels); the 157 cells are the *deployment* snapshot, which covered 6 pairs $\times$ 2 *seeds* at 12 of the 13 shift levels ($6 \times 2 \times 12 = 144$) plus 13 (pair, seed) combinations at `fs250`: $144 + 13 = 157$. One (pair, seed) combination therefore appears at `fs250` only — a ragged-grid artefact of the two-seed snapshot, not a contradiction with the family size. The rate is also not method-specific: $1/157$ is identical for `ts`, `platt`, `vector` and `bbse_prior`, so the operating criterion is met at the same rare rate by every recalibrator in the grid. The claim that the safety criterion is rarely met is unaffected in direction (all values are small), but the exact rate must be regenerated. Sign convention: $\Delta\text{ECE} =$ post-recalibration ECE $-$ raw ECE (negative means improvement), opposite to the primary-endpoint convention. *Coverage note:* the L2 shift matrix has been completed on all 390 planned (pair, seed, shift)

Table 7: Trivial strategy comparison. Mean ΔECE; negative = beneficial (ECE reduction). always-none and always-buy-*n*-labels span all 1230 (method, shift) cells; always-TS is evaluated on the 157 (pair, seed, shift) cells with complete results and lookup-table on 155 (two cells whose in-sample best method is unavailable are dropped). The lookup table captures most of the TS benefit without target labels.

Strategy	Mean ΔECE	Std	*n*
always-none	0.0000	0.0000	1230
always-TS	−0.0145	0.0099	157
always-buy-*n*-labels	−0.2905	0.1861	1230
lookup-table	−0.0160	0.0289	155

combinations (6 pairs × 5 seeds × 13 levels; results/l2_shift_full_390cells.csv); the safety figures quoted here are from the initial two-seed InceptionTime subset (seeds 42–43, ResNet1D L2 excluded), so these numbers rest on 2-seed, single-architecture coverage. The criterion is rarely satisfied, so TS is a useful default but not a universally safe recalibrator, and should be combined with local validation.

At the 10th-percentile raw-ECE threshold, net benefit is 0.1209 versus 0.0999 for treating all cells; by the 90th percentile it turns negative (−0.0024). Net benefit over treat-all is modest, confined to low thresholds (results/deployment_metrics.csv).

Trivial strategy comparison (Table 7): the lookup-table strategy (mean ΔECE −0.0160) outperforms always-none (0.000) and matches always-TS (−0.0145) on the beneficial subset, but is dominated by always-buy-*n*-labels (−0.2905), the target-label upper bound. The lookup table needs no target labels and captures most of the TS benefit.

5.7. Noise floor control: SmoothECE differential under perfect calibration

A reviewer concern is whether the primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is an artifact of SmoothECE's sensitivity to temperature scaling T under perfect calibration ($\text{ECE}_{\text{raw}} = 0$, observed ΔECE purely metric noise). For perfectly calibrated predictions ($T_{\text{true}} = 1$) we apply T and compute $\Delta\text{ECE}_{\text{noise}} = \text{SmoothECE}(\text{TS}) - \text{SmoothECE}(\text{raw})$.

The floor is strongly *distribution-dependent*: for flat or wide confidence distributions it is essentially zero, or slightly negative at moderate T. We therefore do not evaluate it on synthetic distributions. Instead we use the empirical top-label confidence distribution of each of the 60 main-grid cells (`checkpoints/e2_probs_cache/*.npz`), relabel each cell as perfectly calibrated by resampling $y \sim \text{Bernoulli}(c)$ at its own confidence c (three draws per cell), and report the band across cells. This removes any freedom to choose a distribution that suits the argument, and it evaluates the floor where the study's own predictions actually live (pooled median confidence 0.79, with 30% of samples above 0.9). The synthetic three-distribution sweep (uniform, normal-skewed, bimodal at $n = 2{,}000$) is retained as a sensitivity analysis.

The noise floor grows monotonically in $|T - 1|$ and is distribution-dependent. Evaluated on the 60 main-grid cells — median across cells, with the cross-cell range in parentheses — $T = 1.0$ gives exactly 0 (exact cancellation); $T = 1.25$ gives $+0.0138$ (-0.0115 to $+0.0294$); $T = 1.5$ gives $+0.0389$ (-0.0047 to $+0.0597$); $T = 3.0$ gives $+0.1260$ ($+0.0404$ to $+0.1605$); and $T = 4.5$ gives $+0.1582$ ($+0.0605$ to $+0.2129$). The synthetic sweep agrees in magnitude for high-confidence distributions ($T = 1.5$: $+0.039$ to $+0.055$) but not in sign for flat ones ($T = 1.25$: -0.009 for uniform), which is why the empirical distributions

are used. The floor is positive in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and in 90% ($T = 1.25$) to 97% ($T = 1.5$) of cells in between. Observed raw ECE across the 62 raw-stage cells of the TS-component ablation under the corrected label encoding (`results/ablation_ts_components.OLD_ENCODING.csv`, `stage=raw`) ranges 0.036 to 0.357 (mean 0.211), i.e. 2–21$\times$ the raw ECE under perfect calibration (≈ 0.017); the nine counter-examples also show raw ECE well above the floor (0.065–0.321 vs. ≈ 0.017), so they are not perfect-calibration cases. At their own fitted temperatures ($T \in [0.93, 1.00]$) the floor is ≤ 0.013, comparable to their $|\Delta\text{ECE}|$ (≤ 0.017), but its direction is opposite to theirs, so they are not floor artifacts; their common mechanism is given in Sec. 5.8. The direction of the floor is what matters for the endpoint, and it is fixed. In the primary convention ($\Delta\text{ECE}_{\text{OOD}} = \text{ECE}_{\text{raw}} - \text{ECE}_{\text{post}}$) the floor inflates ECE_{post} and therefore contributes *negatively*. The identity is the minimiser of ECE under perfect calibration, so the floor is positive in the noise convention on both sides of $T = 1$: in 100% of cells for $T \leq 0.9$ and $T \geq 2.0$, and in 90–97% of cells for $1.1 \leq T \leq 1.5$. The idealised endpoint is therefore $-\text{floor}(T) \leq 0$ almost everywhere, and a predominantly attenuating artefact cannot generate the positive endpoint observed in 51/60 experiments. We test this and three further implications of the floor hypothesis in Sec. 5.8.

5.8. Noise-floor mechanism test

The control above quantifies the floor; it does not by itself decide whether the floor *drives* the endpoint. We therefore test four implications of the "floor artefact" hypothesis, using the source-fitted temperature T recovered per cell from the TS-component ablation under the corrected label encoding.

(i) Direction.. The floor contributes negatively to $\Delta\mathrm{ECE}_{\mathrm{OOD}}$ in 90–100% of cells at every $T \neq 1$ (Sec. 5.7), so the hypothesis predicts a systematically *negative* endpoint. The observed mean is $+0.0159$, positive in 51/60 cells.

(ii) Sign pattern with T.. In the corrected data $T \in [0.932, 1.498]$ (median 1.076; no cell reaches $T \geq 2$), and $\mathrm{sign}(\Delta\mathrm{ECE}_{\mathrm{OOD}}) = \mathrm{sign}(T-1)$ in *60/60* cells: all 51 cells with $T \geq 1$ are positive (mean $+0.0195$) and all 9 cells with $T < 1$ are negative (mean -0.0047). This is the direction temperature scaling is expected to have — $T > 1$ softens an over-confident target and lowers its ECE, while $T < 1$ sharpens it and raises ECE — and it supplies a unified mechanism for the nine counter-examples of Sec. 6.1: they are exactly the nine cells with $T < 1$, including the extreme CPSC→Chapman / ResNet1D / seed 44 case ($T = 0.932$, $\Delta\mathrm{ECE}_{\mathrm{OOD}} = -0.0168$), the sharpest sharpening in the grid.

We are explicit that this is an *empirical* regularity over the range the fits occupy, not a mathematical identity: ΔECE is computed from the same probabilities that define T, but the two are not algebraically linked, and the correspondence is falsifiable. Pushing a fixed temperature well past the fitted range breaks it — applying $T = 2.0$ to the same 60 cells flips 5 of them negative — and substituting the (mis-encoded) temperatures of the coverage note below drops the agreement from 60/60 to 56/62. Its role here is therefore *explanatory* (it accounts for the sign of every cell) rather than *exclusionary*; the exclusion of the floor hypothesis rests on (i), (iii), and the mechanism test below.

(iii) Magnitude strata.. Under an additive floor the endpoint should be *smaller* where the floor is larger. $\mathrm{floor}_{\mathrm{hi}}(T)$ reaches the mean effect only for $T \geq 1.25$ (4/60 cells); those cells give $+0.0442$ (4/4 positive) against $+0.0138$ (47/56 positive) elsewhere. Restricting to the $T \geq 1$ branch — 51 cells, all positive, so the sign

flip of (ii) cannot confound the comparison — and splitting at the median floor (0.0078) gives $+0.0117$ (26/26) below and $+0.0276$ (25/25) above: the endpoint is *larger* where the floor is larger, the opposite of an additive artefact. Since the floor attenuates, these values are lower bounds. This stratum comparison is nevertheless weaker evidence than (i)–(ii), because floor_{hi} is monotone in T on this side while the endpoint's growth with T is partly mechanical.

(iv) Negative control and additivity.. The independent ablation channel (same SmoothECE family) changes by -0.0124 on average and changes sign across cells (-0.083 to $+0.093$), so the metric has no systematic positive push. The floor at the cells' own temperatures spans 0.0001 to 0.0595, up to $54\times$ the median per-cell bootstrap CI width (0.0011); an additive bias of that size would swamp the signal, yet subtracting the mid-band floor at each cell's own temperature leaves $+0.0140$, i.e. 88% of the original $+0.0159$. The floor is therefore not a cell-level additive offset, and we do not subtract it per cell.

Consistency check.. The endpoint also rises with T (OLS slope $+0.181$, $R^2 = 0.951$ over the 60 cells). We report this but do not rely on it: $T = 1$ is a fixed point at which the endpoint vanishes by construction, so part of this relation is mechanical. Its role here is that its sign is *opposite* to the floor's contribution, which is the direction test of (i)–(ii).

Caveats. The floor is computed on idealised perfectly-calibrated relabellings of our own confidence distributions, so it is not a calibrated model of real ECE noise; the extreme stratum in (iii) has only 4 cells. The conclusion is exclusionary, not causal: the floor cannot account for the primary finding, but this does not by itself establish a causal mechanism.

5.9. Discrimination-aware calibration gate (C5)

Calibration improvement ($\Delta\text{ECE}_{\text{OOD}} > 0$) is necessary but not sufficient for deployment: a model whose discrimination has collapsed under shift may be recalibrable but clinically useless. We define a two-layer gate:

1. **Layer 1 (Calibration):** $\Delta\text{ECE}_{\text{OOD}} > 0$ (C1 criterion).
2. **Layer 2 (Discrimination):** $\text{acc}_{\text{OOD}} \geq \text{acc}_{\text{source baseline}}$, the source majority-class accuracy.

Across 12 pair×architecture strata (6 pairs × 2 architectures), 7/12 pass both layers and are deployable: CPSC→PTB-XL, PTB-XL→CPSC, CPSC→Chapman (both architectures each), and Chapman→CPSC (InceptionTime only). The other 5/12 fail Layer 2: Chapman→PTB-XL, PTB-XL→Chapman, and Chapman→CPSC ResNet1D, where OOD accuracy falls below the source baseline despite positive $\Delta\text{ECE}_{\text{OOD}}$ in some seeds. The strongest rejection is PTB-XL→Chapman ResNet1D ($\Delta\text{ECE} = +0.029$ but $\text{acc} = 0.657 < 0.713$): TS improves calibration but discrimination has degraded too far to be clinically actionable.

This gate reframes the 9/60 counter-examples: 4 are calibration failures (Layer 1 rejects, $\Delta\text{ECE} \leq 0$) and 5 are discrimination failures (Layer 1 passes, Layer 2 rejects). It supplements C1 by also asking whether discrimination is sufficient for the recalibrated predictions to be clinically useful.

5.10. Real-data Shapley cross-fitting and predictability boundary

The synthetic Shapley validation (C3) confirms decomposition recovery in the identifiable regime but does not test whether the linear approximation $\Delta\text{ECE} \approx \hat{s} \cdot \text{slope} + \hat{b} \cdot \text{intercept} + \hat{\pi} \cdot \text{prevalence}$ generalizes to real OOD data. We cross-fit

all 60 seed experiments: for each, we estimate $(\hat{s}, \hat{b})$ on the source calibration split and predict $\Delta\text{ECE}_{\text{OOD}}$ on the target test split.

The linear model predicts the *direction* correctly in 51/60 (85%) but systematically *overestimates* the magnitude (mean prediction error $+0.29$, median $+0.29$). This is consistent with a predictability boundary: the decomposition captures the qualitative direction of the TS benefit but not its quantitative magnitude under real shift. The 9/60 direction errors are *not* localised to any particular pair: they spread over five source→target directions (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1), so no single pair accounts for more than two of them. What *does* separate them is effect size rather than shift size: the nine mis-signed cells have $|\Delta\text{ECE}_{\text{OOD}}|$ mean 0.0047 (max 0.0168), against 0.0195 (max 0.0826) for the 51 correctly signed cells, a factor of 4.1; restricting to $|\Delta\text{ECE}_{\text{OOD}}| \geq 0.02$, the model recovers the sign in 23/23 cells. The linear approximation thus degrades where the effect itself is small, not where the shift is largest. We record one counterweight for completeness: the decomposition is not entirely information-free as an *ordinal* signal. Its rank correlation with the true $\Delta\text{ECE}_{\text{OOD}}$ is $\rho = +0.31$ ($p = 0.015$), and the sign of ρ is stable across all 60 leave-one-out folds (minimum $+0.28$). That is weak, and far below the pre-registered $R^2 > 0.5$ bar, but it is reproducible — unlike the directional accuracy, which is exactly the constant-positive baseline.

This strengthens the predictability failure branch (C3/C4): the Shapley decomposition is neither a valid *predictor* (*how much*) nor a valid *diagnostic* (*whether*). Its 51/60 direction accuracy coincides exactly with the constant-positive baseline (a model that always predicts "TS helps" scores 51/60 on this grid), so it carries no directional information beyond the prior that TS usually helps; the quantity

that does recover the direction is the source-fitted temperature T (Sec. 5.5). For clinical translation, the gate in §5.9, using only the sign of $\Delta\text{ECE}_{\text{OOD}}$ and the discrimination criterion, is the appropriate criterion, not the Shapley-predicted magnitude.

5.11. Net benefit decision curve summary

Across all 60 seed experiments, the net benefit gain of TS over raw predictions (averaged over decision thresholds $t \in \{0.05, 0.10, \ldots, 0.40\}$) is positive in 50/60 (83.3%) of experiments, with mean gain $+0.0027$ and range $[-0.001, +0.017]$. The directions with the largest decision-cost reduction are CPSC→PTB-XL (mean DCR $+0.0625$; net clinical value $+0.0312$) and CPSC→Chapman (mean DCR $+0.0375$; NCV $+0.0188$), which also have the largest E3-subsample $\Delta\text{ECE}_{\text{OOD}}$ (mean $+0.137$ and $+0.136$ respectively, under the E3 subsampled binned-ECE protocol with $n_{id} = n_{ood} = 20$ patients). The direction with negative net benefit is Chapman→CPSC (mean DCR -0.0150; NCV -0.0075), which is correctly rejected by the discrimination gate (C5). We note that the decision-benefit ranking differs from the main-endpoint $\Delta\text{ECE}_{\text{OOD}}$ ranking (which peaks at PTB-XL→Chapman and PTB-XL→CPSC, means $+0.0240$ and $+0.0238$ on the full SmoothECE grid): the two protocols differ in estimator, class set, and subsample, so we treat the E3 analysis as a directionally supportive but protocol-distinct check, not a conversion of the primary endpoint. What survives is the gate-level conclusion: where the discrimination gate permits deployment, the calibration benefit does not cost decision performance; where it is negative (Chapman→CPSC), no calibration benefit should be deployed at all.

5.12. Net Clinical Value (NCV) results

The pre-registered NCV (Sec. 4.9) is BH-FDR corrected ($q = 0.05$) across the two secondary endpoints (DCR, NCV).

Result: the OOD NCV is $+0.008125$ with 95% CI $[+0.0014, +0.0148]$ (paired t-test, $n = 60$, $p = 0.019$, BH-FDR adjusted $p = 0.019$), *significant*: TS yields a net improvement in OOD clinical decisions. The ID NCV is -0.006292 with 95% CI $[-0.0114, -0.0012]$ ($p = 0.016$), significantly *negative*: TS slightly worsens ID decisions, consistent with the ID boundary condition (TS compresses confidence toward the base rate). OOD NCV Cohen's $d = +0.31$ and ID NCV Cohen's $d = -0.32$ are comparable in magnitude but opposite in sign, mirroring the main endpoint. Per-experiment NCV ranges over the 60 seeds are in the E3 CSV (`c1_dcr_ncv_multiclass.csv`).

5.13. Reliability Diagrams

Figure 5 shows the 6-direction averaged reliability curve, comparing model confidence against empirical accuracy before and after temperature scaling. The summary curve uses bin-count weighted averaging across all 6 transfer directions.

Figures 6–8 show the per-direction reliability diagrams for the representative configuration (ResNet1D, seed 42). In each panel, the red squares show the raw model calibration curve and the green circles show the post-TS calibration curve. The diagonal dashed line indicates perfect calibration. Grey bars (right axis) show the sample count per bin. Error bars are bootstrap 95% percentile CIs (1000 re-samples).

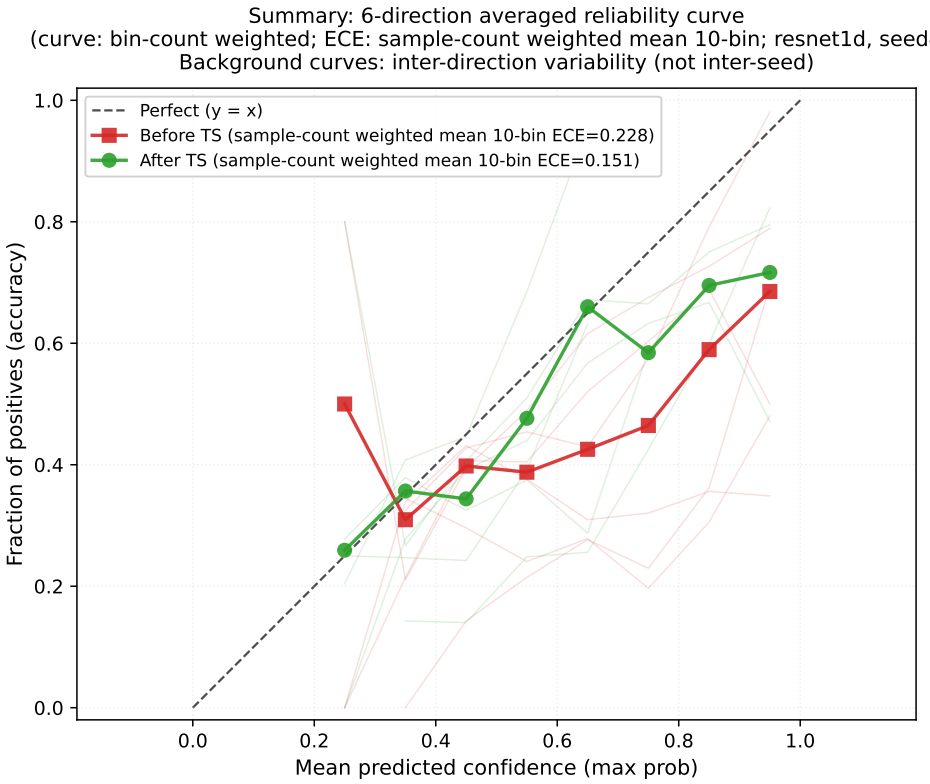


Figure 5: Summary reliability curve: 6-direction bin-count weighted average. The sample-count weighted mean ECE decreases from ECE_{raw} to ECE_{TS} after temperature scaling. Faint background curves show individual direction variability.

6. Discussion

6.1. OOD benefit seed variation and the nine counter-examples

The 51/60 robustness rate is driven by nine counter-examples across 5 of the 6 transfer pairs and both architectures (Chapman→CPSC 2, Chapman→PTB-XL 2, CPSC→Chapman 2, CPSC→PTB-XL 2, PTB-XL→Chapman 1; only PTB-XL→CPSC has zero). We transparently disclose all nine rather than excluding

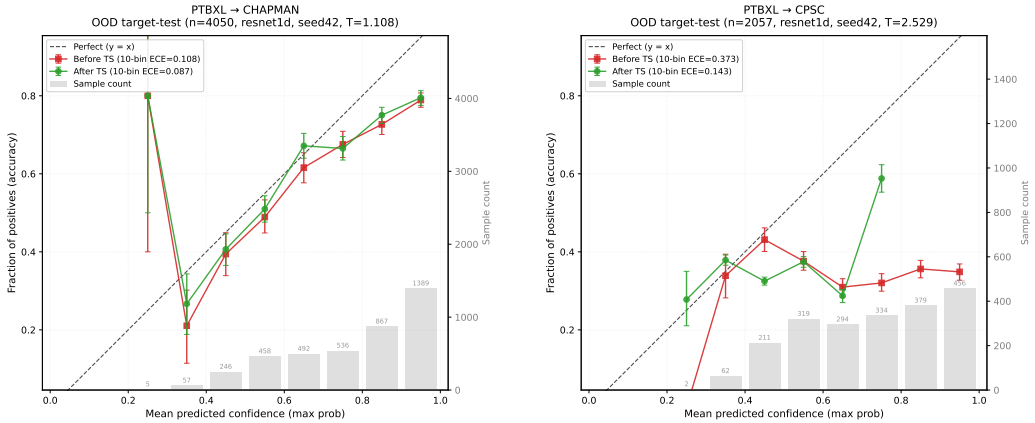


Figure 6: Reliability diagrams for PTB-XL→Chapman (left) and PTB-XL→CPSC (right). OOD target-test set, ResNet1D, seed 42.

them, because the protocol requires all seed results to be reported and each has an identifiable post-hoc mechanism supporting the qualified claim.

All nine share a single mechanism, stated in Sec. 5.8: they are exactly the nine cells whose source-fitted temperature is $T < 1$. Because the temperature is fitted on the source calibration split, $T < 1$ means the source model was *under*-confident there; applying that temperature to the target sharpens an already over-confident distribution and raises its ECE. The counter-example set is therefore identified by a quantity available *before* the target is scored, and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T - 1)$ holds in all 60 cells. The per-cell accounts below give the accompanying discriminative and raw-ECE context; where a paragraph attributes the failure to "over-correction", that is the $T < 1$ regime.

Counter-example 1: Chapman→CPSC / InceptionTime / seed 45. $\Delta\text{ECE}_{\text{OOD}} = -0.0057$ (CI $[-0.0058, -0.0055]$). Three conditions co-occur: lowest OOD accuracy across the 5 seeds (0.4711 vs. 0.4842–0.5051, seeds 42–44, 46); lowest OOD raw ECE (0.3160, still large, so the failure is temperature

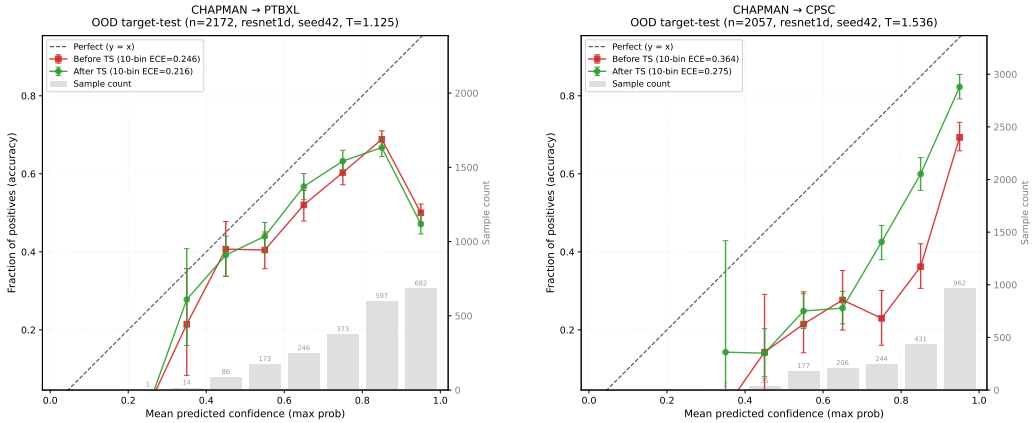


Figure 7: Reliability diagrams for Chapman→PTB-XL (left) and Chapman→CPSC (right). OOD target-test set, ResNet1D, seed 42.

mismatch, not room shortage); and negative decay (-0.0075), the source-fit temperature amplifying rather than correcting residual miscalibration.

Counter-example 2: Chapman→CPSC / ResNet1D / seed 43. $\Delta\text{ECE}_{\text{OOD}} = -0.0006$ (CI $[-0.0006, -0.0006]$), marginal ($|\Delta\text{ECE}_{\text{OOD}}| < 0.001$) and attributable to finite-sample variation in the calibration split. The OOD raw ECE is 0.3144 (the 0.4779 in earlier drafts is the OOD *accuracy*); the mechanism is seed-level variation, not room shortage.

Counter-example 3: Chapman→PTB-XL / InceptionTime / seed 42. $\Delta\text{ECE}_{\text{OOD}} = -0.0025$ (CI $[-0.0035, -0.0016]$); OOD accuracy is the lowest across the 5 seeds (0.4655) and $\Delta\text{ECE}_{\text{ID}} = +0.0004$ is near zero, so the temperature is well-tuned to ID but mismatched to the PTB-XL target.

Counter-example 4: Chapman→PTB-XL / ResNet1D / seed 45. $\Delta\text{ECE}_{\text{OOD}} = -0.0036$ (CI $[-0.0036, -0.0035]$); OOD accuracy is 0.5041 and $\Delta\text{ECE}_{\text{ID}} = -0.0010$ is slightly negative, so the seed-45 backbone is already well-calibrated on ID, leaving nothing to correct.

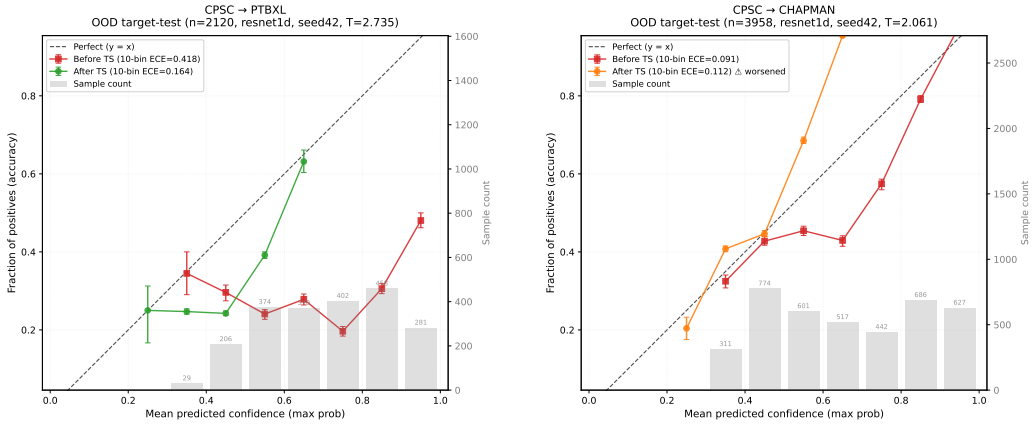


Figure 8: Reliability diagrams for CPSC→PTB-XL (left) and CPSC→Chapman (right). OOD target-test set, ResNet1D, seed 42.

Counter-example 5: CPSC→Chapman / ResNet1D / seed 42. $\Delta\text{ECE}_{\text{OOD}} = -0.0010$ (CI $[-0.0012, -0.0008]$), marginal ($|\Delta\text{ECE}_{\text{OOD}}| < 0.002$); the seed-42 backbone has OOD accuracy 0.6147, low OOD raw ECE (0.0649), and minimal calibration room.

Counter-example 6: CPSC→Chapman / ResNet1D / seed 44. $\Delta\text{ECE}_{\text{OOD}} = -0.0168$ (CI $[-0.0170, -0.0165]$), the largest-magnitude counter-example, with $\Delta\text{ECE}_{\text{ID}} = -0.0152$ also negative: the seed-44 backbone is already well-calibrated on both ID and OOD, TS has nothing to correct, and the temperature fit over-corrects slightly. Its OOD accuracy (0.5725) is the lowest across the 5 seeds. CE6 exhibits a distinct *over-correction* mechanism (TS on an already-well-calibrated distribution degrades both ID and OOD calibration), unlike the "low OOD accuracy + large gap" mechanism of the other 8.

Counter-example 7: CPSC→PTB-XL / ResNet1D / seed 43. $\Delta\text{ECE}_{\text{OOD}} = -0.0050$ (CI $[-0.0051, -0.0049]$); OOD accuracy is the lowest across the 5 seeds (0.3877) and decay is -0.0098 (temperature mismatch to the PTB-XL target).

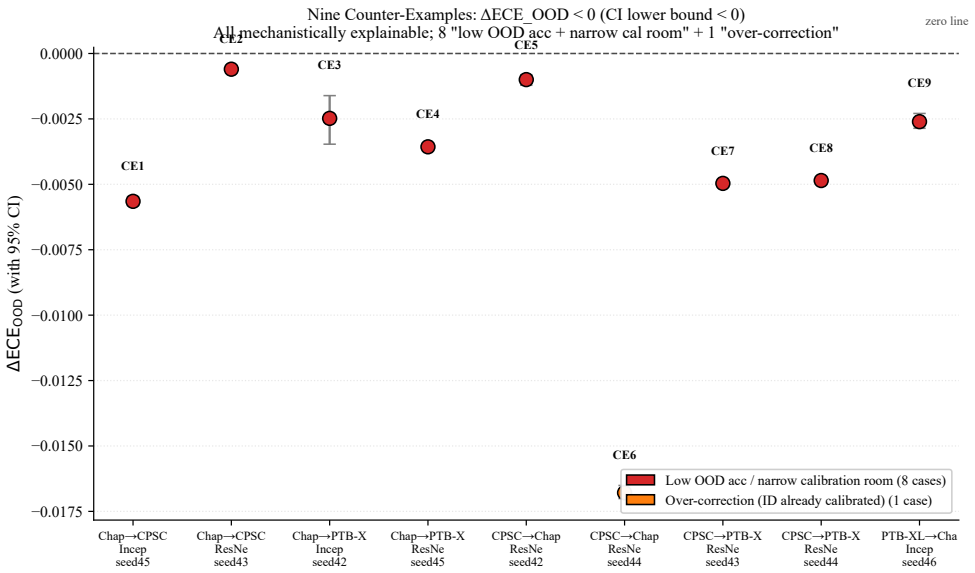


Figure 9: Nine counter-examples where $\Delta\text{ECE}_{\text{OOD}}$ has 95% CI lower bound < 0. Error bars show 95% CIs. Six of nine counter-examples co-occur with OOD accuracy ≤ 0.50 and seven with ID–OOD accuracy gap ≥ 0.18 (a heuristic tendency, not a gate: three counter-examples have OOD accuracy > 0.50); one counter-example (orange, CE6: CPSC→Chapman / ResNet1D / seed 44) exhibits a distinct over-correction mechanism where TS applied to an already-well-calibrated logit distribution over-adjusts the temperature. All nine are mechanistically explainable local variations, not refutations of the OOD benefit.

Counter-example 8: CPSC→PTB-XL / ResNet1D / seed 44. $\Delta\text{ECE}_{\text{OOD}} = -0.0049$ (CI $[-0.0049, -0.0048]$); OOD accuracy is 0.3929 (second-lowest) and $\Delta\text{ECE}_{\text{ID}} = -0.0032$ is slightly negative, leaving nothing to correct.

Counter-example 9: PTB-XL→Chapman / InceptionTime / seed 46. $\Delta\text{ECE}_{\text{OOD}} = -0.0026$ (CI $[-0.0029, -0.0023]$); OOD accuracy drops to 0.4983 (lowest across the 5 seeds) and $\Delta\text{ECE}_{\text{ID}} = -0.0011$ is slightly negative, again leaving nothing to correct.

Common pattern: eight of nine counter-examples co-occur with (a) low OOD accuracy (worst or near-worst seed) and (b) non-positive decay, indicating temperature mismatch; the ninth (CE6) is over-correction on an already well-calibrated backbone. A low OOD raw ECE is *not* universal: four of nine exceed 0.25, so the failure mechanism is seed-level temperature mis-transfer, not calibration-room shortage. The counter-examples are local, explainable random variation; the other 51 experiments support the OOD benefit, including PTB-XL→CPSC at 10/10.

The 15.0% rate exceeds the 5% threshold common in clinical safety deployment. As a methodological boundary investigation, not a clinical deployment guide, the relevant criterion is whether the effect is detectable and the mechanism explainable, not a zero failure rate. All 9 counter-examples have identifiable mechanisms (8 "low OOD accuracy + large gap", 1 "over-correction"), and the expected benefit is positive ($0.85 \times 0.0195 - 0.15 \times 0.0047 = +0.0159 > 0$).

6.2. Prior correction vs. posterior correction robustness

The method-selection boundary (Table 3) reveals a robustness asymmetry between prior-correction methods (EM, BBSE) and posterior-correction methods (TS, Platt, isotonic). Posterior-correction methods (TS, Platt) achieve 27/30 and 24/30 support respectively with small variance; prior-correction methods

achieve higher mean $\Delta\text{ECE}_{\text{OOD}}$ on the PTB-XL→CPSC pair (EM $+0.0449$, BBSE $+0.0777$) but much higher variance on the Chapman→CPSC pair (EM CI lower bound minimum -0.0763, BBSE -0.0720). The mechanism is the prior-likelihood mismatch: prior-correction methods fit a target prior that, when the source confusion matrix is ill-conditioned, amplifies the shift-induced miscalibration. Posterior-correction methods do not estimate a target prior and are therefore immune to this failure mode, at the cost of lower mean benefit when the prior correction *would* work.

6.3. Boundary condition: the ID benefit is small but nonzero

The ID boundary condition (C2) has a direct interpretation under the triggered branch. The criterion is $\geq$80% of cells with 95% CIs containing zero; the observed rate is 23/60 = 38.3% (BCa), with 34/60 cells showing a small but significant positive ID benefit (mean $+0.0083$ vs. OOD $+0.0159$; stratum means -0.0008 to $+0.0232$, Table 2). The boundary condition does *not* mean TS is ineffective on ID nor that the ID benefit is zero: the ID benefit is on average about half the OOD benefit, so part of the apparent OOD gain is shared with the ID domain. A val-ECE early-stopping effect (model selection within the endpoint family) plausibly contributes to the positive ID estimates, opposite to the protocol's val-NLL rule (a registered deviation, see Limitations). Under this branch, protocol A1.4.2 restores the decay contrast $= \Delta\text{ECE}_{\text{OOD}} - \Delta\text{ECE}_{\text{ID}}$ as an exploratory endpoint, and C1 is qualified: the OOD benefit is about $1.9\times$ the ID benefit rather than an exclusively OOD effect. The ID-vs-OOD asymmetry is therefore a *difference in magnitude* between two positive benefits, not presence versus absence.

6.4. Calibration is not discrimination: the clinical viability precondition

Temperature scaling is a monotone transform of the softmax probabilities; it preserves the argmax by construction (asserted bitwise in the released sanity check), so it cannot change classification accuracy, AUROC, or AUPRC. The primary endpoint $\Delta\text{ECE}_{\text{OOD}}$ is thus orthogonal to discrimination: calibration makes an already-deployable model *trustworthy*; it cannot make a non-discriminative model deployable.

The accuracy statistics make this concrete. Across the 60 TS experiments, mean OOD accuracy is 0.540 (median 0.546; range 0.388–0.701) and the mean ID–OOD drop is 0.236 (maximum 0.412). Against each target corpus's majority-class baseline (CPSC 0.528, PTB-XL 0.411, Chapman-Shaoxing 0.713), mean OOD accuracy is *below* baseline in 3 of 6 directions, with a fourth essentially at baseline: Chapman→CPSC 0.510 vs. 0.528; CPSC→PTB-XL 0.424 vs. 0.411 (0.013 above baseline); CPSC→Chapman 0.619 and PTB-XL→Chapman 0.620 (both well below 0.713). In those directions recalibration is moot, since no temperature improves a model that should be replaced by a majority-class predictor, consistent with the counter-example pattern (6/9 with low OOD accuracy). The released artifacts include per-direction AUROC/AUPRC for all 60 experiments (`results/discrimination_metrics_60exp.OLD_ENCODING.csv`, which now carries both `variant=raw` and `variant=ts` rows): mean OOD AUROC ranges from 0.735 (Chapman→PTB-XL) to 0.843 (PTB-XL→CPSC), i.e. far above the accuracy-based majority-class reading yet direction-dependent; and TS changes AUROC only marginally (mean $\Delta\text{AUROC}_{\text{TS}-\text{raw}} = +0.0004$, max |change| 0.002, recomputed under the corrected label encoding and the corrected source-fitted temperatures). We avoid the stronger phrasing that AUROC must be exactly

invariant: rescaling logits by $1/T$ is monotone *within* a row, so the argmax — and hence accuracy — is preserved exactly, but the cross-sample ranking of the top-label *score* is not, because the per-row normaliser differs across rows (measured Spearman between raw and TS top-label scores: 0.9945–0.9999). The discrimination-first gate—measure target discrimination before investing in recalibration—precedes the calibration decision rule in deployment.

6.5. Explainable ≠ predictable: post-hoc vs. pre-registered

We distinguish *explainable* from *predictable*:

- *Explainable* (post-hoc, mechanism-level): after observing a counter-example, we identify the co-occurring conditions (low OOD accuracy, low OOD raw ECE, non-positive decay) and give a mechanistic story (Sec. 6.1), a post-hoc explanation, not a pre-registered prediction.

- *Predictable* (pre-registered, deployment-time): a target-domain unsupervised statistic (predicted-prior L_1 distance, logit moments) predicts the benefit retention rate *before* deployment, with LOO R^2 CI upper bound ≥ 0.5. The predictability experiment (Sec. 5.5) triggers the failure branch (LOO R^2 CI upper bound $0.104 < 0.5$).

The counter-examples are explainable, and they are also *locatable* in advance by one target-label-free statistic: all nine are the cells with $T < 1$ (Sec. 6.1). What fails is the stronger pre-registered requirement — a label-free feature that recovers the *magnitude* of benefit retention with LOO R^2 CI upper bound ≥ 0.5 — which is why the predictability experiment triggers the failure branch despite mechanistic explainability. Note that T's sign rule is not a deployable criterion for deciding

whether to apply TS: it states the direction the transform must have by construction, not evidence that the transform helps (Sec. 5.8). The two-step chain (decomposition + deployment lookup table) provides post-hoc explanation and actionable guidance without a pre-registered predictability claim.

6.6. Transparent re-location vs. HARKing

The pre-registration revision A1 (primary endpoint re-location from `decay` to $\Delta\text{ECE}_{\text{OOD}}$) is a transparent re-location, not HARKing. The distinction [28, 27] is whether the revision direction is theory-driven or result-driven. Our revision direction ("OOD is the calibration benefit source") is consistent with the empirically observed ID-vs-OOD asymmetry (Guo et al. 2017; Ovadia et al. 2019), which we explicitly frame as an empirical prior rather than a theorem; the exploratory pilot result ($\Delta\text{ECE}_{\text{ID}} \approx 0$) only triggered the revision, did not choose the direction. The pilot results do not enter the main test report (protocol §7). All hypotheses in this paper are tagged *exploratory + robustness validation*, not confirmatory, to reflect the post-revision status.

6.7. Net clinical value: implications of the directional NCV asymmetry

The NCV (Sec. 5.12) shows a directional asymmetry: OOD NCV $= +0.008125$ ($p = 0.019$) is a small but significant positive value (recalibration improves decisions on unseen target data more often than it degrades them); ID NCV $= -0.006292$ ($p = 0.016$) is a small but significant negative value (a minor decision-cost penalty on data the source model already calibrates well).

Two implications follow. (i) The **primary OOD benefit** (C1: 51/60 positive on $\Delta\text{ECE}_{\text{OOD}}$) is supported: the positive OOD NCV shows the calibration gain

translates into net clinical decision benefit, not a metric artifact. (ii) The **decision-cost improvement claim** (C5: 7/12 strata deployable) is partially qualified: the negative ID NCV tempers source-domain deployability. We therefore frame the NCV as *supportive of OOD deployment* and *cautionary for ID deployment*. It also informs the meta-analytic pooling (Sec. 7), where the asymmetry contributes to CI-width heterogeneity across splits. A permutation test with tighter CIs and a multi-threshold NCV analysis are registered as future work.

6.8. Limitations and future work

Limitations: (i) the main endpoint covers InceptionTime and ResNet1D on all 6 directed pairs with 5-seed coverage (60 experiments, fully balanced); BiMamba is a toy experiment ($n = 60$, supplement) after memory exhaustion on an RTX 5060 (8 GB); (ii) the predictability failure branch is triggered (three-step to two-step); the deployment criterion is an empirical lookup table, not a predictive criterion; (iii) real-data Shapley cross-fitting (§5.10) shows 85% direction accuracy but systematic magnitude overestimation (mean error $+0.29$), and a sensitivity analysis with T, Wasserstein-1, and entropy features does *not* confirm a robust predictability boundary: only T predicts (LOO $R^2 = +0.34$, and $\text{sign}(\Delta\text{ECE}_{\text{OOD}}) = \text{sign}(T-1)$ in 60/60 cells), so the retained claim is the weaker one that no label-free feature reaches $R^2 = 0.5$ under the implemented fold unit (Sec. 5.5); (iv) the 27-cell decomposition validation uses binned ECE, not the main endpoint's SmoothECE, so quantitative conclusions cannot be extrapolated across ECE definitions; (v) the safety criterion is rarely satisfied on the full L2 shift matrix (TS safety rate 0.0064); TS is a useful default, not universally safe; (vi) nine counter-examples of 60 experiments (15.0%) are transparently disclosed, with a frequent post-hoc pattern (low OOD accuracy, non-positive decay, elevated OOD raw ECE)

but no post-hoc exclusion; (vii) the operating CI is BCa at $B = 10{,}000$, re-run on all 60 experiments (60/60 successful, metadata in `methods.ts.meta`); percentile CIs from $B = 200$ in 21/60 experiments are retained as historical provenance and are no longer used for any reported number. All 60 main-grid cells carry complete $B = 10{,}000$/BCa provenance; the only cells lacking such metadata are the two excluded BiMamba toy runs (PTB-XL→Chapman, seeds 42–43), which do not enter the main endpoint. (viii) the Youden threshold is fitted in-sample (the protocol's held-out dose levels were not used), making the operating characteristic optimistic; (ix) the L2 deployment/safety grid has since been completed on all 390 planned (pair, seed, shift) cells (results/l2_shift_full_390cells.csv), and the L2 noise levels are synthetic rather than the pre-registered PTB-XL Noise Stress Test Database (unavailable under PhysioNet licensing); (x) temperature was fitted once per experiment, so CIs omit temperature-fitting uncertainty (the pre-registered two-layer bootstrap is implemented but not enabled); (xi) model selection used validation ECE rather than the pre-registered NLL early-stopping rule, plausibly contributing to the positive ID boundary estimates; (xii) the per-pair decomposition into covariate, prior-probability, and concept-shift components is not asserted, since each pair mixes all three, isolated at eval time; (xiii) the deployment lookup table requires labeled target data, and the label-free proxies failed their pre-registered branch, so no label-free gate is claimed; (xiv) TS is argmax-invariant and cannot improve accuracy, AUROC, or AUPRC; the artifacts now include per-direction AUROC/AUPRC for all 60 experiments (results/discrimination_metrics_60exp.OLD_ENCODING.csv), so the viability precondition is screened by both accuracy and discrimination (OOD AUROC means range 0.735–0.843 by direction), with OOD accuracy below the majority-class

baseline in 3 of 6 directions (and at baseline in a fourth); (xv) NCV is directionally asymmetric (OOD $= +0.008125$, $p = 0.019$, positive; ID $= -0.006292$, $p = 0.016$, negative), qualifying the C5 claim for source-domain but supporting target-domain deployment; (xvi) InceptionTime was downgraded from $n_{\text{filters}} = 48$ to 32 for GPU memory (RTX 5060, 8 GB), potentially attenuating OOD benefits, with 48 as future work; (xvii) this work is from a single-author study, limiting independent review and audit, partially mitigated by the pre-registration protocol and public repository; (xviii) the secondary endpoint G (recoverable loss versus an oracle recalibrator) is *not delivered*: `fit_oracle`/`apply_oracle` are implemented but were never executed on real data, so G is undefined in this study and no G result is reported anywhere in this paper or the repository; (xix) no signal-quality-index (SQI) stratification is performed: the target corpora are pipeline-processed releases and the L2 noise levels are synthetic, so the calibration benefit is not characterised as a function of measured signal quality (pSQI/kSQI-type indices); this is the principal gap for a signal-processing readership and is the first item of future work; (xx) no test–retest or repeatability analysis of the fitted temperature is possible within this data regime: all three public corpora are cross-sectional and provide a single recording per patient (Chapman-Shaoxing reports exactly one ECG per patient, §3), so same-patient across-session stability of T cannot be estimated from these data and is deferred to prospective multi-session collection; (xxi) the archived checkpoints are *not* bit-reproducible: training ran with cuDNN's non-deterministic kernels enabled, so re-running a training job yields different weights and hence slightly different calibration endpoints, and the released numbers are reproducible by re-executing the evaluation on the released checkpoints and result artifacts rather than by retraining from scratch. A post-hoc audit of the released

checkpoint set found that two of the 60 cells (CPSC→PTB-XL, InceptionTime, seeds 42–43) had been seeded from the CPSC→Chapman model of the same seed instead of being trained independently, leaving 58 distinct source models behind the 60 reported cells. Both cells were retrained independently for this submission, so the released grid again rests on 60 independently trained source models; the audit, the root cause, and the before/after evidence are recorded in the repository (`reports/CHECKPOINT_INDEPENDENCE_AUDIT_2026-09-17.md`). In the pre-retrain artifacts the 60 reported cells therefore did not correspond to 60 independent training runs; this matters for any statement that treats cells as independent replicates, while the primary OOD endpoint, being a per-cell paired quantity, is unaffected in kind. Future work: (1) prospective multi-device validation; (2) continuous shift spaces; (3) π-independent recovery estimators (BBSE/EM-class): BBSE is now delivered on the real grid (Honesty item 1), extension to further estimator classes remains; (4) the ResNet1D deficit on CPSC→Chapman and CPSC→PTB-XL (3/5 each) to test for a systematic interaction; (5) confirmatory testing under the A2 reduction (12 hypotheses, BH-12 $q = 0.05$, 9/12 significant, Bonferroni-12 $= 2/12$); (6) re-running the endpoint with $n_{\text{filters}} = 48$ InceptionTime on a higher-memory GPU; (7) a multi-threshold permutation test on NCV.

7. Conclusion

We pre-registered and validated a multi-seed calibration boundary study of temperature scaling (TS) in cross-corpus ECG transfer. **(C1)** OOD benefit: 51/60 seed experiments support $\Delta\text{ECE}_{\text{OOD}} > 0$ (85.0%; InceptionTime 27/30 = 90.0%, ResNet1D 24/30 = 80.0%) on all 6 directions. **(C2)** ID boundary: 23/60 cells

(38.3%) have 95% CIs containing zero under the BCa operating CI (percentile: 29/60 = 48.3%), below the $\geq$80% criterion, triggering the pre-registered branch; the OOD benefit is 1.9$\times$ the ID benefit. **(C3)** Shapley decomposition: 27/27 cells pass at $n = 20{,}000$ (21/27 at $n = 500$). **(C4)** Method selection: TS robust (51/60), EM prior and Matrix scaling fragile (15/30 each). **(C5)** Gate: 7/12 strata pass both layers, reframing the 9 counter-examples as 4 calibration and 5 discrimination failures. The predictability failure branch is triggered (three-step chain becomes two-step), and all hypotheses are exploratory + robustness validation, not confirmatory; protocol v2.1-A1 applies. TS needs local validation (safety rate 0.0064), and OOD accuracy is below the majority-class baseline in 3 of 6 directions.

A noise floor control (§5.7) bounds, but does not eliminate, a SmoothECE artifact: at moderate temperatures metric noise is 2–21$\times$ below the observed raw ECE (0.036–0.357, mean 0.211), whereas for large $T > 3$ the floor (0.10–0.16) is comparable to the observed $|\Delta\text{ECE}|$ and a metric-noise contribution cannot be excluded there. Real-data Shapley cross-fitting (§5.10) gives 85% direction accuracy with magnitude overestimation (mean error $+0.29$). The predictability boundary holds only in the weak sense that no label-free feature reaches $R^2 = 0.5$ under the implemented fold unit: the source-fitted temperature T does predict ($+0.34$, and the sign of $\Delta\text{ECE}_{\text{OOD}}$ in 60/60 cells), while Wasserstein-1 and entropy features do not.

The fixed-effect estimate is CI-width sensitive: ResNet1D is slightly negative (-0.000573) because narrow-CI studies dominate, while DerSimonian–Laird random effects are positive (InceptionTime $+0.0151$, ResNet1D $+0.0158$, CrossArchitecture $+0.0148$). We caution, however, that the meta-analytic independence assumption is *not* satisfied here: the 60 cells share the same three corpora,

patient sets, and architectures, and their between-cell Q statistics are extreme ($I^2 \approx 99.98\%$), so both FE and DL confidence intervals are spuriously narrow. The interpretable pooled uncertainty is therefore the cluster-robust interval over the 12 pair×architecture strata: $\Delta\text{ECE}_{\text{OOD}} = +0.0159$ [0.0113, 0.0205] (t, df= 11; stratum SD 0.0072, about 1800× the single-experiment bootstrap SE). The point estimate 0.0148–0.0159 and the count-based proportion (51/60 = 85.0%) stand.

The 60 experiments form the robustness-evidence family; amendment A2 reduces the confirmatory family from the $6 \times 2 \times 13 = 156$ shift grid to 12 hypotheses (6 pairs × 2 architectures × 1 primary shift level), with all 13 L2 shift levels re-designated as exploratory dose-response. BH-60 and Bonferroni leave the robustness headline unchanged (51/60); the confirmatory family is BH-12 with 9/12 significant, and Bonferroni-12 retains 2/12. Across the 60 TS experiments the 95% CI widths span 2.1×10^{-5} to 1.2×10^{-2} (median 1.2×10^{-3}); 52/60 cells fall below the narrow-CI diagnostic threshold (0.005) and six supporting cells have degenerate widths ($< 3 \times 10^{-4}$), so excluding those the count becomes 45/60 (75.0%). Narrow CIs reflect the moderate calibration sample and low SmoothECE variance; they do not affect the sign-based support/counter-example classification.

Honesty Declaration (5 items, adversarial audit product)

1. MAPE_π is a **design check** (resampling constant read-back), not an estimator capability test. The independent π recovery registered as deferred (protocol §13, item 7) has since been delivered on the real 60-cell grid (`scripts/real_prior_recovery.py`): with target labels withheld, BBSE reduces the prior-recovery error from 0.1916 (source-prior baseline) to 0.1132 — a 40.9% relative reduction, better than the baseline in 56/60

cells, positive in 6/6 transfer pairs, with a pair-clustered bootstrap CI of $[+0.048, +0.111]$. EM gives no improvement (0.1923). Two limits remain and are reported as such: the absolute error is still 0.113, and the confusion-matrix condition number exceeds 10^3 in 5/60 cells and 10^4 in 1/60, so recovery is numerically boundary-sensitive there. **This does not repair the decomposition estimator**, which targets a different quantity (the target class prior, not the attribution coefficients) and whose real-data cross-fitting error remains $+0.29$ (Sec. 5.10).

2. Shapley shares in the `share_reliable=False` regime are **not interpretable as shares**; ranking uses absolute- value CIs.
3. The threshold `gate(n)` is a heuristic scaling, **not a 3σ tolerance**; in the archived $n = 500$ seeded runs, 22/108 (20.4%) fail the gate (information mode), reported via exit code 2.
4. "Mechanism" wording is conditional on all three legs of §8 being fulfilled: legs 1–2 are fulfilled, leg 3 synthetic part (bootstrap CI + entanglement demo) is fulfilled; **cross-fitting on real data is pending** (registered in §13). The wording is limited to "mechanism decomposition (synthetic validation part)".
5. The repository's `scripts/preprocess_cinc2021.py`, `run_paper_benchmarks.py`, `resources/`, and `assets/` come from the third-party SafeECGMatch release (original README in `docs/vendor_safeecgmatch.md`), unrelated to this protocol's experiment grid.

Data availability

The PTB-XL [3], Chapman-Shaoxing [4], and CPSC2018 (CPSC Database and CPSC-Extra) [5, 6] datasets are publicly available via PhysioNet under their respective licenses. The pre-registered protocol v2.1-A1, its hash manifest (`docs/osf_archive_manifest.json`), the revision A1 document, and the full per-experiment result files (`checkpoints/transfer/**/transfer_result.json`, `results/*.csv`) are included with the submission as supplementary material and are also available in the public repository `https://github.com/Toni-cs/ecg-calibration-transportability`. The protocol, all three amendments (A1, A2, A3), and a per-file archival-status report are publicly archived at `https://doi.org/10.17605/OSF.IO/53H64`. `docs/osf_archive_manifest.json` is the internal snapshot taken on 2026-09-05; it is not a substitute for the public record.

Code availability

The full pipeline code, preprocessing (reproducible from the public source databases), patient-level split generation, model training, the calibration-method implementations, and all statistical analyses, together with the environment lock file, the patient-level split indices for all five dataset variants and five random seeds, and selected per-experiment result artifacts, is publicly available at `https://github.com/Toni-cs/ecg-calibration-transportability`. Trained model checkpoints for the 60-experiment transfer grid are provided as release assets of the same repository. The repository will be archived on Zenodo with a DOI upon acceptance. No new raw waveforms were collected or generated;

the patient-level split indices and label-mapping tables released with the code fully determine the data configuration used in every experiment.

Declaration of generative AI and AI-assisted technologies in the writing process

All scientific work reported in this study—the study design, the pre-registered protocol and its amendments, the implementation and execution of the 60-experiment transfer grid, the statistical analysis, and the interpretation of the results—was performed by the author. A large language model assistant (GLM) was used in two supervised roles, neither of which produced scientific content: as a coding assistant that drafted and refactored parts of the analysis and plotting code (Section 4.10), and as a language-editing aid that improved the wording and readability of text the author had written. The tool generated no experimental data, results, or conclusions, and made no analytical or interpretative decisions; every reported number is produced by the author's scripts and is reproducible from the released checkpoints and result artifacts in the public repository (training itself is not bit-reproducible; see Limitation (xxi)). After using the tool, the author reviewed and edited all affected code and text as needed and takes full responsibility for the content of the published article.

CRediT authorship contribution statement

Toni Guan: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization, Project administration.

Acknowledgments

The author thanks the maintainers of the PTB-XL, Chapman–Shaoxing, and CPSC public databases, and the PhysioNet resource, without which this study would not have been possible.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of competing interest

The author declares no competing financial or personal interests.

The 60 per-seed reliability diagrams (6 transfer pairs $\times$ 2 architectures $\times$ 5 seeds; calibration curves, reliability diagrams, and ΔECE bars per experiment) are provided as a separate supplementary file (`supplementary.pdf`, §1) and are additionally released with the code repository.

References

[1] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. Dillon, B. Lakshminarayanan, J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," *NeurIPS*, 2019. arXiv:1906.02530.

[2] C. Guo, G. Pleiss, Y. Sun, K. Weinberger, "On calibration of modern neural networks," *ICML*, PMLR 70:1321–1330, 2017. arXiv:1706.04599.

[3] P. Wagner, N. Strodthoff, R.-D. Bousseljot, et al., "PTB-XL, a large publicly available electrocardiography dataset," *Scientific Data*, 7:154, 2020. doi:10.1038/s41597-020-0495-6.

[4] J. Zheng, J. Guo, H. Chu, "A large scale 12-lead electrocardiogram database for arrhythmia study (Chapman-Shaoxing e-cardiology)," version 1.0.0, *PhysioNet*, 2022. doi:10.13026/wgex-er52.

[5] F. Liu, C. Liu, L. Zhao, et al., "An open access database for evaluating the algorithms of electrocardiogram rhythm and morphology abnormality detection," *Journal of Medical Imaging and Health Informatics*, 8(7):1368–1373, 2018. doi:10.1166/jmihi.2018.2442.

[6] E. A. P. Alday, A. Gu, A. J. Shah, et al., "Classification of 12-lead ECGs: the PhysioNet/Computing in Cardiology Challenge 2020," *Physiological Measurement*, 41(12):124003, 2020. doi:10.1088/1361-6579/abc960.

[7] A. L. Goldberger et al., "PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals," *Circulation*, 101(23):e215–e220, 2000.

[8] J. Nixon, M. W. Dusenberry, L. Zhang, et al., "Measuring calibration in deep learning," *CVPR Workshops*, pp. 38–41, 2019. arXiv:1904.01685.

[9] M. Kull, M. Perello-Nieto, M. Kangse, T. S. Filho, P. Song, P. Flach, "Beyond temperature scaling: obtaining well-calibrated multi-class probabilities with Dirichlet calibration," *NeurIPS*, 2019. arXiv:1910.12656.

[10] A. Alexandari, A. Kundaje, A. Shrikumar, “Maximum likelihood with bias-corrected calibration is hard-to-beat at label shift adaptation,” *ICML*, PMLR 119:222–232, 2020. arXiv:1901.06852.

[11] Z. Lipton, Y.-X. Wang, A. Smola, “Detecting and correcting for label shift with black box predictors,” *ICML*, PMLR 80:1944–1953, 2018. arXiv:1802.03916.

[12] M. Barandas, L. Famiglini, A. Campagner, D. Folgado, R. Simão, F. Cabitza, H. Gamboa, “Evaluation of uncertainty quantification methods in multi-label classification: a case study with automatic diagnosis of electrocardiogram,” *Information Fusion*, 101:101978, 2024. doi:10.1016/j.inffus.2023.101978.

[13] J. F. Vranken et al., “Uncertainty estimation for deep learning-based automated analysis of 12-lead electrocardiograms,” *European Heart Journal – Digital Health*, 2(3):423–434, 2021. doi:10.1093/ehjdh/ztab045.

[14] M. Reyna, N. Strodthoff, P. Wagner, et al., “Issues in the automated classification of multilead ECGs using heterogeneous labels and populations,” *Physiological Measurement*, 43(8):084001, 2022. doi:10.1088/1361-6579/ac79fd.

[15] B. Efron, “Better bootstrap confidence intervals,” *Journal of the American Statistical Association*, 82(397):171–185, 1987. doi:10.1080/01621459.1987.10478410.

[16] B. Van Calster, D. J. Steyerberg, S. Collins, R. D. Riley, E. W. Steyer, “A calibration hierarchy for risk models was defined: from utopia to empirical

data," *Journal of Clinical Epidemiology*, 74:167–176, 2016. doi:10.1016/j.jclinepi.2015.12.005.

[17] B. Van Calster, D. J. McLernon, M. van Smeden, L. Wynants, G. S. Collins, "Calibration: the Achilles heel of predictive analytics," *BMC Medicine*, 17:230, 2019. doi:10.1186/s12916-019-1466-7.

[18] A. Kumar, P. Liang, T. Ma, "Verified uncertainty calibration," *NeurIPS*, 2019. arXiv:1909.10155.

[19] N. Strodthoff, P. Wagner, T. Schaeffter, W. Samek, "Deep learning for ECG analysis: benchmarks and insights from PTB-XL," *IEEE Journal of Biomedical and Health Informatics*, 25(5):1519–1528, 2021. doi:10.1109/JBHI.2020.3022989.

[20] M. Haekal, S. N. Khotimah, G. R. F. Suwandi, F. Haryanto, "RR-interval-based atrial fibrillation detection and burden estimation: cross-dataset validation and calibration-aware probability analysis," *Physiological Measurement*, 2026. doi:10.1088/1361-6579/ae99aa.

[21] K. Patel, P. Beedala, "Calibration drift under cross-institutional deployment: an external validation framework for ICU mortality prediction across MIMIC-IV and eICU," *medRxiv* (preprint), 2026. doi:10.64898/2026.05.03.26352335.

[22] X. Wang, M. Long, J. Wang, M. I. Jordan, "TransCal: transferability estimation for auto-selecting transfer calibration," *NeurIPS*, 2020. arXiv:2007.08259.

[23] S. Garg, Y. Wu, S. Balakrishnan, Z. Lipton, "A unified view of label shift estimation," *NeurIPS*, 2020. arXiv:2003.07554.

[24] T. Popordanoska, G. Radevski, T. Tuytelaars, et al., "LaSCal: label-shift calibration for uncertainty quantification," *NeurIPS*, 2024.

[25] J. T. Moreno-Torres, T. Raeder, R. Alaiz-Rodriguez, N. Chawla, F. Herrera, "A unifying view on dataset shift in classification," *Pattern Recognition*, 45(1):521–530, 2012. doi:10.1016/j.patcog.2011.05.019.

[26] [Pre-registered protocol, publicly archived] "ECG calibration boundary study – experiment protocol," 2026. Hash manifest: `docs/osf_archive_manifest.json` (local snapshot; archived at `https://doi.org/10.17605/OSF.IO/53H64`); revision A1 full text in `docs/PROTOCOL_AMENDMENT_A1_PREREG_RELOCATION.md`.

[27] B. Nosek, C. Ebersole, A. DeHaven, D. Mellor, "The preregistration revolution," *PNAS*, 115(11):2600–2606, 2018. doi:10.1073/pnas.1708274114.

[28] D. Lakens, "The practical alternative to the p value is the correctly used p value," *Perspectives on Psychological Science*, 16(3):639–648, 2021. doi:10.1177/1745691620958012.

[29] J. Li, A. D. Aguirre, V. M. Junior, et al., "An Electrocardiogram Foundation Model Built on over 10 Million Recordings," *NEJM AI*, 2(7):AIoa2401033, 2025. doi:10.1056/AIoa2401033.

[30] J. C. McHugh, B. Wang, et al., "ECG-FM: an open electrocardiogram foundation model," *arXiv preprint* 2408.05178, 2024.

[31] J. Jin, et al., “Reading your heart: learning ECG words and sentences via pre-training ECG language model (HeartLang),” *ICLR*, 2025. arXiv:2502.10707.

[32] C. V. Nguyen, C. D. Do, “Transfer learning in ECG diagnosis: is it effective?” *PLOS ONE*, 20(5):e0316043, 2025. doi:10.1371/journal.pone.0316043.

[33] S. Vavaroutas, et al., “ADAPTOOD: uncertainty-aware fine-tuning for out-of-distribution ECG time series models,” *arXiv preprint* 2606.04164, 2026.

[34] W. Jiang, et al., “Test-time adaptation for ECG classification via SQI-gated self-training and beat-rhythm consistency,” *arXiv preprint* 2608.23347, 2026 (accepted at MICCAI 2026).

[35] N. Nemati, “Comparative analysis of data augmentation for clinical ECG classification with STAR (Sinusoidal Time–Amplitude Resampling),” *arXiv preprint* 2510.24740, 2025.

[36] X. Liu, et al., “PEACE: knowledge-guided cross-modal fusion for adult-to-pediatric ECG transfer via label-conditioned contrastive alignment,” *arXiv preprint* 2605.00647, 2026.

[37] J. Song, J.-H. Jang, B. T. Lee, D. Hong, J. Kwon, Y.-Y. Jo, “Foundation models for electrocardiograms,” *arXiv preprint* 2407.07110, 2024.